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AGRICULTURAL & BIOSYSTEMS POWER ENGINEERING

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A heat engine that burns fuel and air inside a combustion chamber located within the engine proper. A machine use to power tractors, combine harvesters, and other self-propelled equipment.

- 
- a. Heat engine
 - b. Internal combustion engine
 - c. External combustion engine
 - d. Rotary engine



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HEAT ENGINE – An engine that uses heat energy from fuel to generate mechanical power for any useful purposes such as driving machines and for electrical power generation.

Major Classification

1. **Internal** – It burns fuel and air inside the cylinder of the engine
2. **External** – It burns the fuel and air outside the cylinder of the engine

Uses of ICE

1. provides power for **mobile equipment** such as tractors, power tillers, and self-propelled equipment
2. provides power for **stationary equipment** such as pumps
3. provides power for **generators** for the production of electricity in the farm



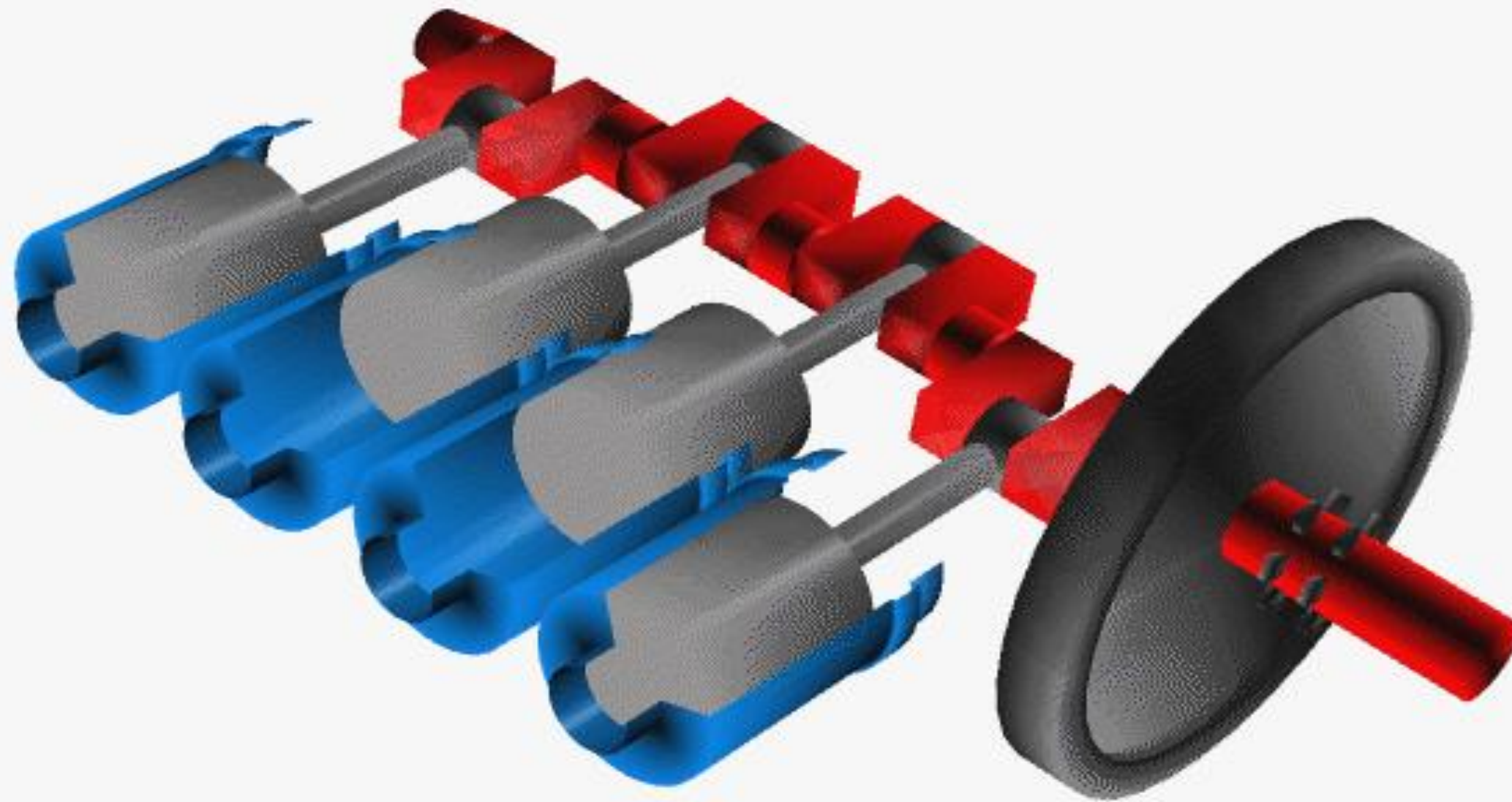
What is the primary function of the crankshaft in an engine?

- a. converts the reciprocating motion of the piston to rotary motion
 - b. connects the piston from the crankshaft
 - c. reduce the sound produced from the engine during combustion of fuel
 - d. automatically controls engine speed
- 



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On the figure are:

1. Crankshaft (red)
2. Flywheel (black)
3. Piston and Connecting Rod (Gray)
4. Cylinder block (blue)



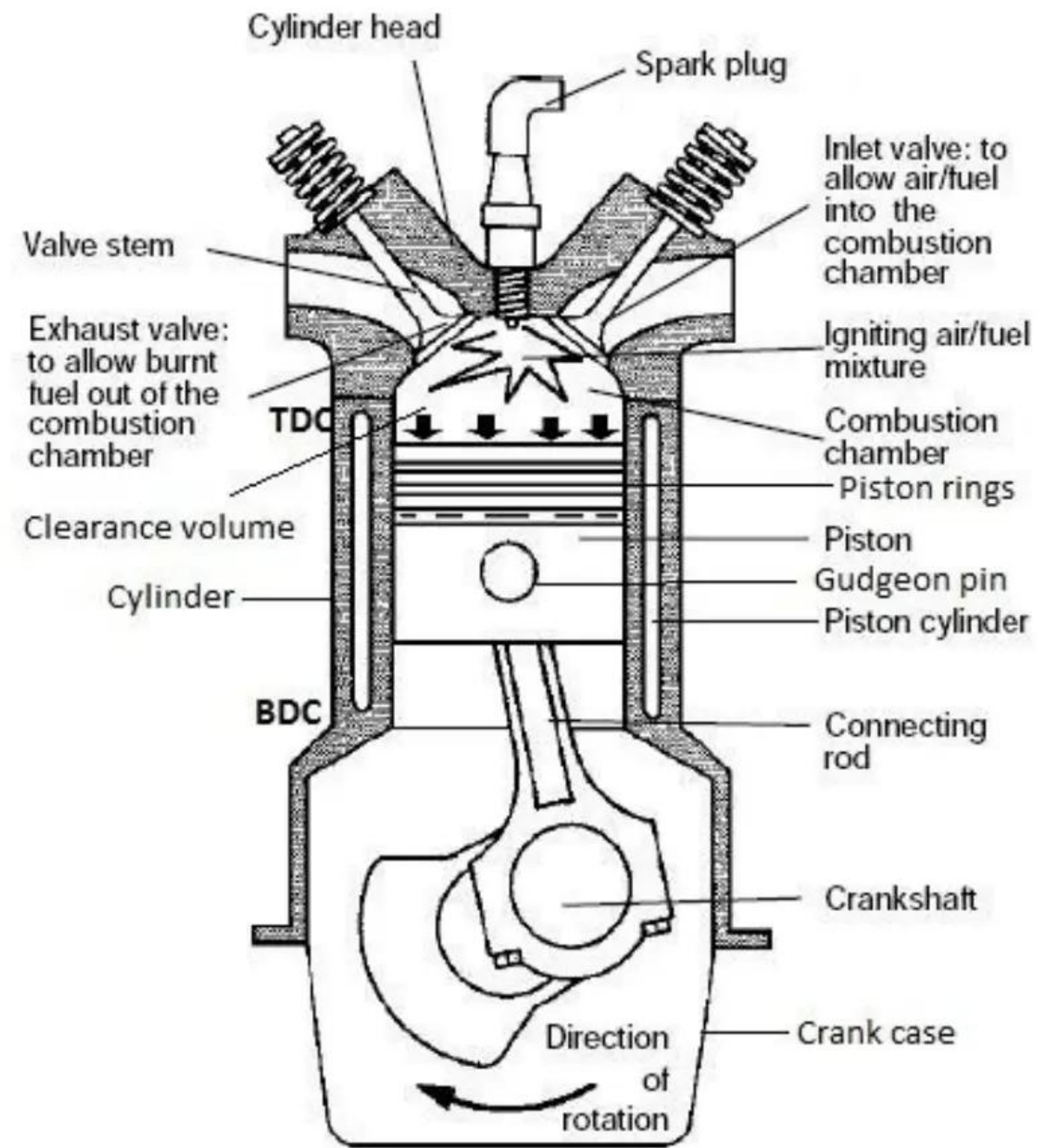
Cylinder head serves as cover of the cylinder block, and it houses the following components EXCEPT _____.

- a. Valves
 - b. Spark plug
 - c. Fuel injector
 - d. None of the above
- 



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It provides the mounting for various components such as inlet and outlet exhaust valves and ducts, spark plugs, fuel injectors, and camshafts. In addition, it gives way to coolant, oil, and combustion gases.

Basic Components of ICE

1. **Cylinder Head** – It houses the valves, spark plug or fuel injector, and serves as cover of the cylinder block
2. **Cylinder Block** – It encloses the piston and confines the expanding gases
3. **Valves** – It allows the flow of fuel, air, or exhaust gases
4. **Piston** – It is a hollow cylinder with closed top upon which the expanding gases act and transmit force to the crankshaft
5. **Crankshaft** – converts the reciprocating motion of the piston to rotary motion
6. **Connecting rod** – connects the piston from the crankshaft
7. **Piston ring** – lubricate and seal the piston (oil ring and compression ring)

8. **Spark plug or fuel injector** – provides ignition for gasoline engine or spray fuel for diesel engine

9. **Crankcase** – encloses the crankshaft and the gear drives

10. **Oil pan** – contains the oil for lubricating the engine

11. **Dip stick** – used for measuring the amount of oil in the crankcase

12. **Governor** – automatically controls engine speed

13. **Flywheel** – stores energy created by the piston needed for successive rotation of the crankshaft

14. **Carburetor** – mixes the right amount of fuel and air

15. **Fuel Tanks** – serves as fuel storage

16. **Muffler** – reduce the sound produced from the engine during combustion of fuel



Arrangement of valves wherein the intake and exhaust valves are located in the cylinder head.

- 
- a. Overhead valve
 - b. I-head arrangement
 - c. L-head arrangement
 - d. Two of the choices above



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- a. Overhead valve
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 - d. Two of the choices above



- **Overhead valve (OHV) or I-head arrangement**
- arrangement of valves wherein the intake and exhaust valves are located in the cylinder head
- **Side valves (SV) or L-head arrangement**
- arrangement of valves wherein the intake and exhaust valves are located on one side of the cylinder block



It is a series of events occurring one after the other in a definite order and repeats itself after the last event has occurred.

- a. Order
- b. Cycle
- c. Relapse
- d. sequence



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- b. Cycle**
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All ICE during their operation follow a definite cycle.

Cycle – It is a series of events occurring one after the other in a definite order and repeats itself after the last event has occurred.

Events for One Cycle

- a. **Intake** – air or air-fuel mixture enters the combustion chamber
- b. **Compression** – air or air-fuel mixture is compressed inside the combustion chamber
- c. **Power** – air-fuel mixture is ignited resulting to a very rapid expansion of the burning mixture
- d. **Exhaust** – burned gases are removed from the combustion chamber, and the cycle is repeated



It is the movement of the piston from one extreme position to the other extreme position to the other extreme position.

- A. Stroke
- B. Revolution
- C. Cycle
- D. Clearance



It is the movement of the piston from one extreme position to the other extreme position to the other extreme position.

A. Stroke

B. Revolution

C. Cycle

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Stroke

- the movement of the piston from one extreme position to the other extreme position.

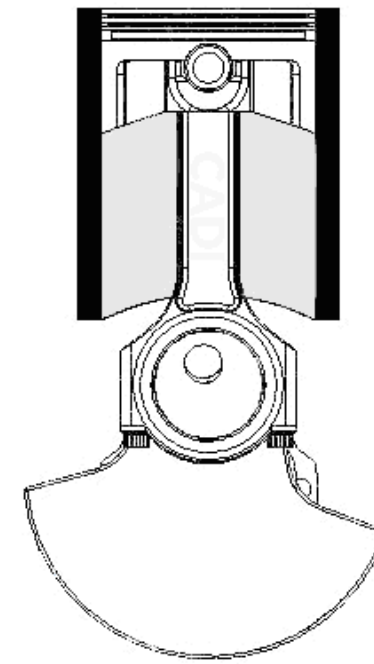
Top dead center position (TDC)

- The **highest point** the piston can reach inside the cylinder.
- Occurs at the end of the **compression stroke** and **exhaust stroke**.

Bottom dead center position (BDC)

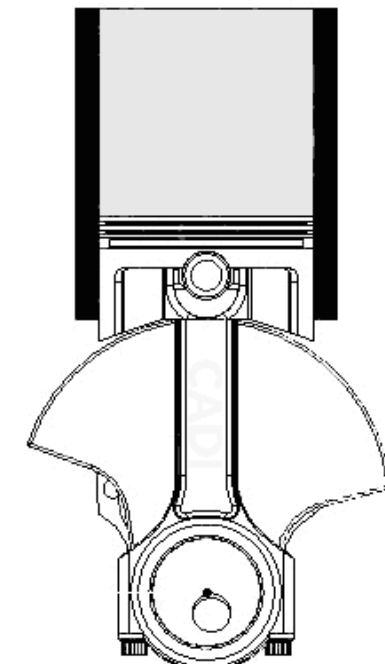
- The **lowest point** the piston can reach inside the cylinder.
- Occurs at the end of the **power stroke** and **intake stroke**.

TOP DEAD CENTER
(TDC)



PISTON IS AT THE
HIGHEST POINT

BOTTOM DEAD CENTER
(BDC)



PISTON IS AT THE
LOWEST POINT



Which is the correct sequence of events in a two-stroke engine cycle?

- A. Intake & Compression → Power & Exhaust
- B. Intake & Power → Compression & Exhaust
- C. Compression & Intake → Exhaust & Power
- D. Compression & Power → Intake & Exhaust



Which is the correct sequence of events in a two-stroke engine cycle?

- A. Intake & Compression → Power & Exhaust
- B. Intake & Power → Compression & Exhaust
- C. Compression & Intake → Exhaust & Power
- D. Compression & Power → Intake & Exhaust

Two-stroke cycle engine

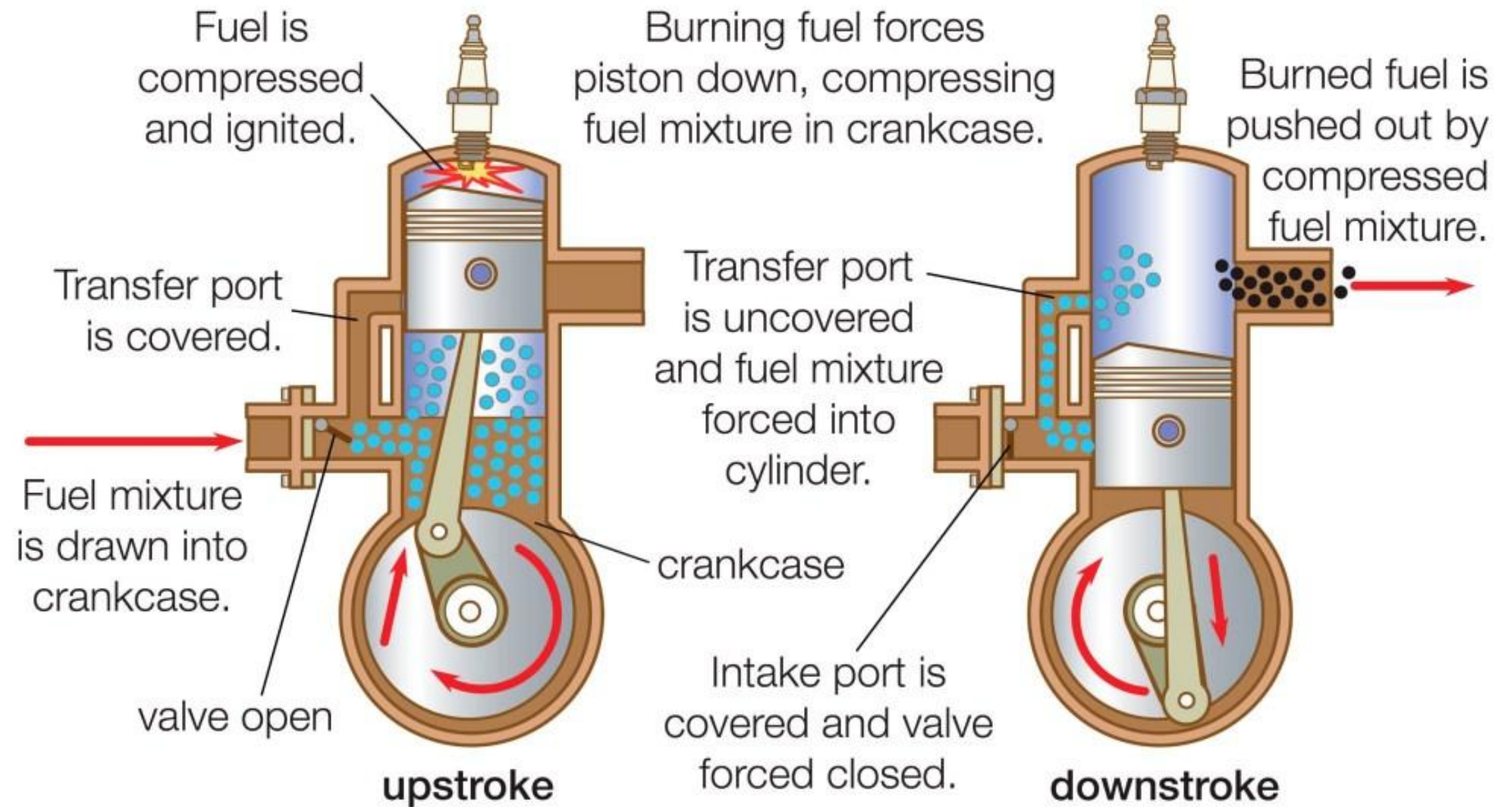
- Requires only two movements of the piston to complete the cycle
- In one stroke, several events have to occur

Upward stroke

- Intake & Compression

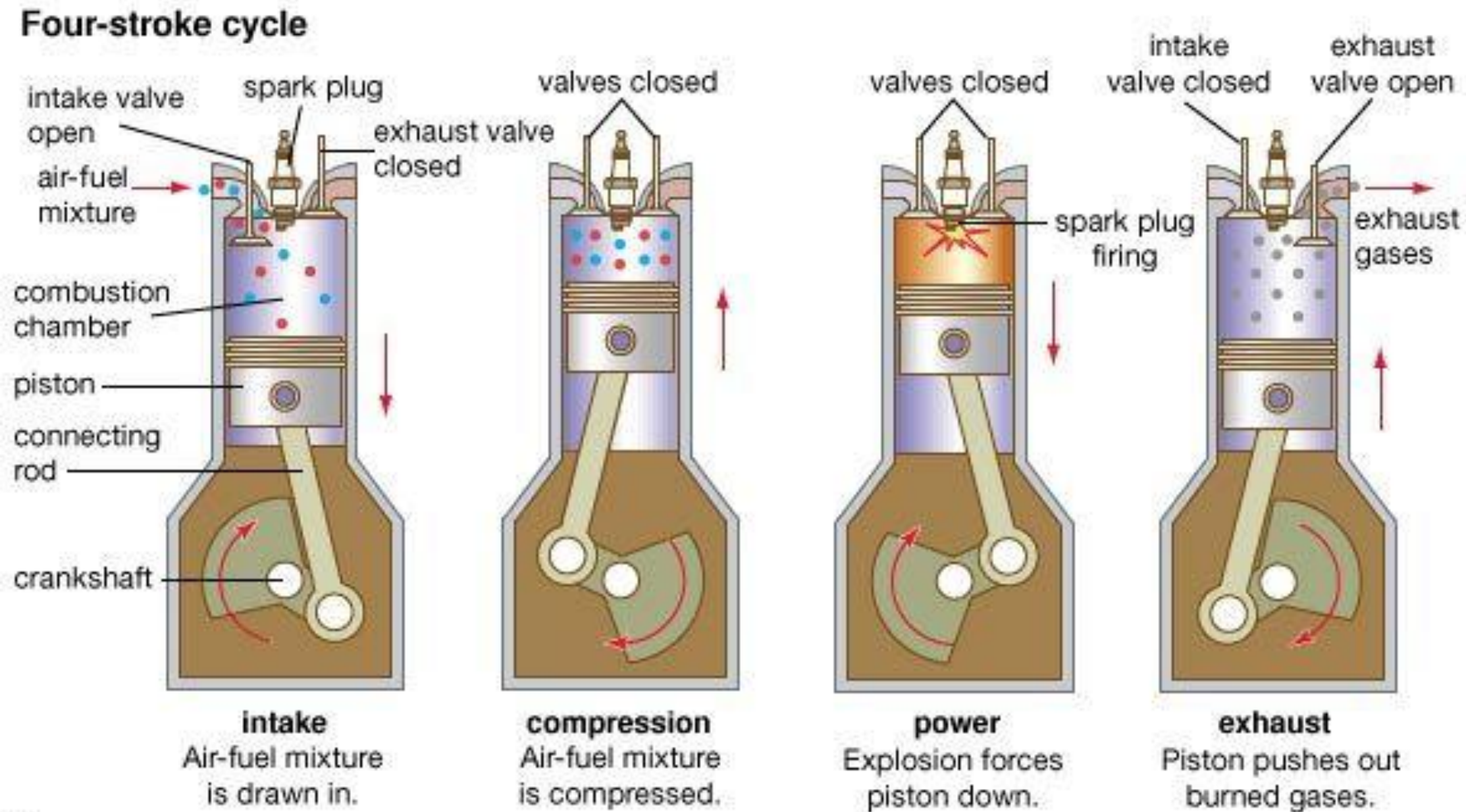
Downward stroke

- Power & Exhaust



Four-stroke cycle engine

- Requires four strokes of the piston to complete the cycle



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Refers to the space inside the cylinder when the piston is at top dead center position.

- a. Piston displacement
- b. Clearance volume
- c. Cylinder volume
- d. Compression ratio



Refers to the space inside the cylinder when the piston is at top dead center position.

- a. Piston displacement
- b. Clearance volume**
- c. Cylinder volume
- d. Compression ratio

Physical Nomenclature

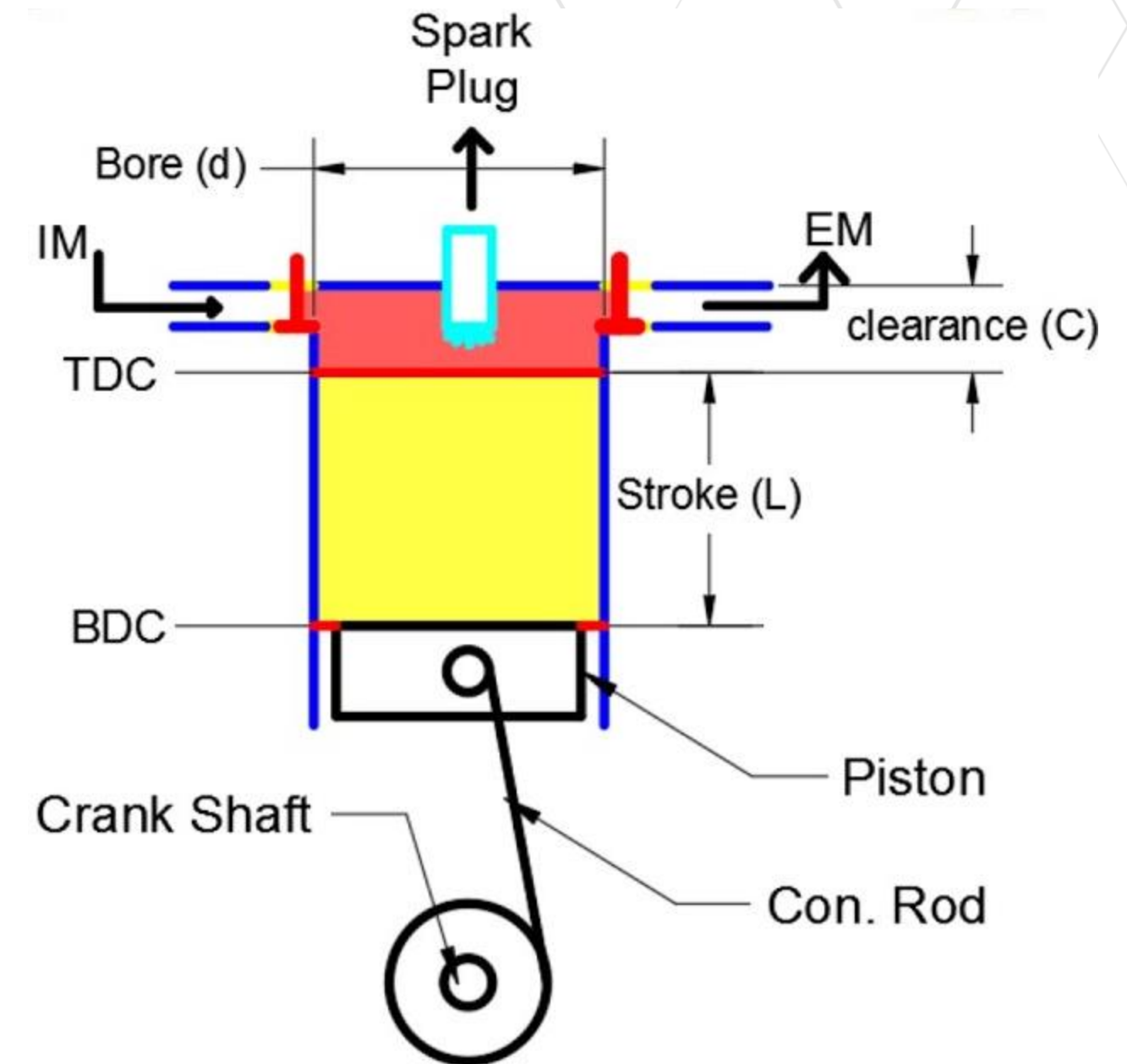
- a. **bore (d)** – the diameter of the engine cylinder
- b. **stroke (l)** – distance the piston travels in one direction
- c. **piston displacement (Pd)** – volume displaced by the piston in one stroke

$$PD = \frac{\pi(bore)^2}{4} \times \text{Stroke}$$

- d. **clearance volume (Cv)** – the space inside the cylinder when the piston is at top dead center position
- e. **cylinder volume** – the sum of piston displacement and clearance volume, Pd + Cv
- f. **compression ratio (Cr)** – ratio of cylinder volume to the clearance volume

$$\text{Compression Ratio} = \frac{\text{Total Volume}}{CV} = \frac{PD + CV}{CV}$$

- g. **engine efficiency** – a measure of the ability of an engine to convert fuel into useful work





Determine the compression ratio of the engine with the following specification: Total volume = 70 cc, Bore = 4 cm and stroke = 5 cm.

- a. 9.76:1
- b. 8.13:1
- c. 7.90:1
- d. 6.02:1

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- a. 9.76:1
- b. 8.13:1
- c. 7.90:1
- d. 6.02:1

$$\begin{aligned} PD &= \frac{\pi(bore)^2}{4} \times \text{Stroke} = \frac{\pi(4 \text{ cm})^2}{4} \times 5 \text{ cm} = 62.83 \text{ cc} \\ CV &= TV - PD = 70 \text{ cc} - 62.83 \text{ cc} = 7.17 \text{ cc} \\ CR &= \frac{TV}{CV} = \frac{70 \text{ cc}}{7.17 \text{ cc}} = \frac{9.76}{1} = 9.76:1 \end{aligned}$$



Determine the bore of the engine if its clearance volume is 7.18 cc, compression ratio is 8:1 and stroke is 4 cm.

- a. 3 cm
- b. 4 cm
- c. 5 cm
- d. 6 cm



Determine the bore of the engine if its clearance volume is 7.18 cc, compression ratio is 8:1 and stroke is 4 cm.

- a. 3 cm
- b. 4 cm**
- c. 5 cm
- d. 6 cm

$$CR = \frac{TV}{CV}$$

$$8 = \frac{TV}{7.18}$$

$$TV = 50.26 \text{ cc}$$

$$TV = PD + CV$$

$$PD = TV - CV = 57.44 \text{ cc} - 7.18 \text{ cc} = 50.26 \text{ cc}$$

$$PD = \frac{\pi(bore)^2}{4} \times \text{Stroke}$$

$$50.26 = \frac{\pi(Bore)^2}{4} \times 4 \text{ cm}$$

$$\mathbf{Bore = 4 \text{ cm}}$$



What is the number of explosion of piston on a single cylinder, four stroke cycle engine, in 300 revolution of the crankshaft?

- a. 100
 - b. 150
 - c. 300
 - d. 600
- 



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- The crankshaft speed directly determines how often the cylinders fire.
- The camshaft's role is more about valve timing and less about the frequency of combustion events.

$$\text{Camshaft RPM} = \frac{\text{Crankshaft RPM}}{2}$$

For 4-stroke:

$$\text{Number of explosions} = \frac{\text{Crankshaft revolutions}}{2}$$

$$\text{Number of explosions} = \text{Camshaft revolution}$$

For 2-stroke:

$$\text{Crankshaft Speed} = \text{Camshaft Speed}$$



A V-12 4-stroke cycle engine, having a directly meshed gears, has a camshaft speed of 2000 rpm. What is the number of explosions per minute this engine can produce?

- a. 8,000
 - b. 24,000
 - c. 36,000
 - d. 48,000
- 



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A V-12 4-stroke cycle engine, having a directly meshed gears, has a camshaft speed of 2000 rpm. What is the number of explosions per minute this engine can produce?

Explosion per minute = camshaft x no. of cylinder

Explosion per minute = 2000 x 12

Explosion per minute = 24,000 rpm



Firing order is the sequence of movement of the piston as it produces power. What is the common firing order for 6 cylinders?

- 
- a. 2 – 4 – 6 – 1 – 3 – 5
 - b. 1 – 2 – 4 – 5 – 3 – 6
 - c. 2 – 1 – 6 – 4 – 5 – 3
 - d. 1 – 5 – 3 – 6 – 2 – 4



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Firing Order

- The firing order 1 – 5 – 3 – 6 – 2 – 4 provides an ideal balance of smoothness, power delivery, and reduced vibrations, making it a common choice for 6-cylinder engines.

No. of Cylinder	Order of Firing
2	1 – 2
3	1 – 2 – 3
4	1 – 2 – 4 – 3 and 1 – 3 – 4 – 2
6	1 – 5 – 3 – 6 – 2 – 4



Why Firing Orders Can Vary:

1. **Engine Design:** Some engines are designed for different purposes (e.g., performance vs. fuel efficiency), which can affect the firing order.
2. **Vibration and Balance:** Some firing orders minimize engine vibrations and improve smoothness.
3. **Manufacturer Preference:** Different manufacturers may design their engines with unique firing orders.



A V-6 4-stroke cycle engine, having a directly meshed gears, has a camshaft speed of 5000 rpm. What is the firing interval?

- a. 20 degrees
 - b. 40 degrees
 - c. 60 degrees
 - d. 120 degrees
- 


$$\text{Firing Interval} = \frac{180^\circ \times \text{No. of stroke}}{\text{No. of cylinder}}$$

A V-6 4-stroke cycle engine, having a directly meshed gears, has a camshaft speed of 5000 rpm. What is the firing interval?

$$\text{Firing Interval} = \frac{180^\circ \times \text{No.of stroke}}{\text{No.of cylinder}}$$

$$\text{Firing Interval} = \frac{180^\circ \times 4}{6}$$

$$\text{Firing Interval} = 120^\circ$$

This means one cylinder fires every 120 degrees of crankshaft rotation.



A V-6 4-stroke cycle engine, having a directly meshed gears, has a camshaft speed of 5000 rpm. What is the firing interval?

- a. 20 degrees
 - b. 40 degrees
 - c. 60 degrees
 - d. 120 degrees
- 



A manufacturer specifies that an engine has a power output of 50 horsepower. This 50 horsepower is referred to as the engine's ____.

- 
- a. Brake horsepower
 - b. Rated horsepower
 - c. Indicated horsepower
 - d. Friction horsepower



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Basic Engine Terminology

- **Indicated Horsepower** – the power generated at the combustion chamber
- **Brake Horsepower** – power available at the drive shaft or crankshaft of the engine
- **Friction horsepower** – power loss due to friction of the piston and other component parts for the machine
- **Rated horsepower** – power specified by the manufacturer



If the power generated inside the combustion chamber of an engine is measured, this is called the ____.

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Intake air pressure increased with the compressor driven off of the engine crankshaft.

- 
- a. Naturally aspirated
 - b. Super charged
 - c. Turbo charged
 - d. Crankcase compressed



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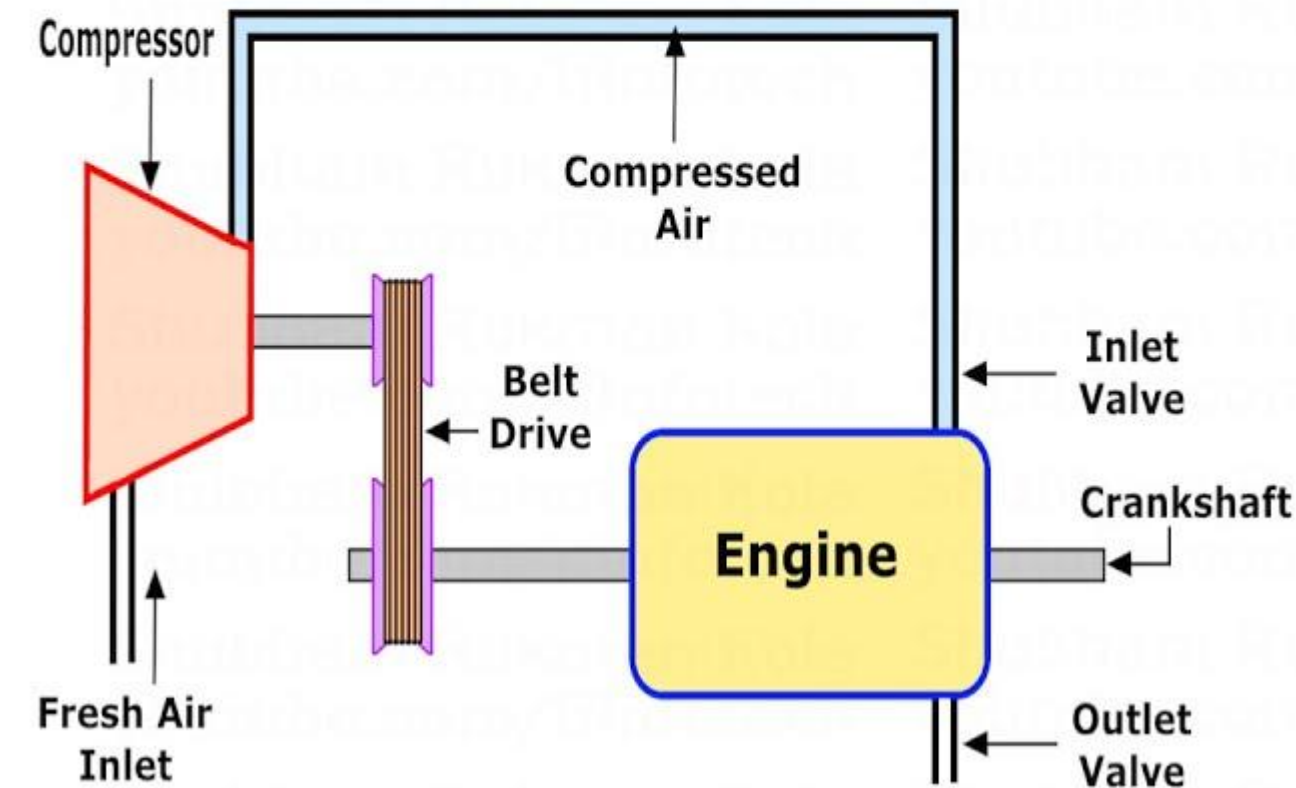
1. Supercharger

- A **supercharger** is an air compressor driven by the **engine's crankshaft**. Its purpose is to force more air into the engine, allowing it to burn more fuel and produce more power.

Process:

1. **Crankshaft Power** - The supercharger is mechanically connected to the engine's crankshaft via a **belt, gear, or chain**.
2. **Compression** - As the crankshaft turns, it spins the supercharger's rotors or impellers, compressing the air.
3. **Increased Air Density** - Compressed air is fed into the engine's intake manifold, increasing oxygen levels.
4. **More Fuel + Air** - With more air in the combustion chamber, the engine can inject more fuel, resulting in greater power output.

Supercharger



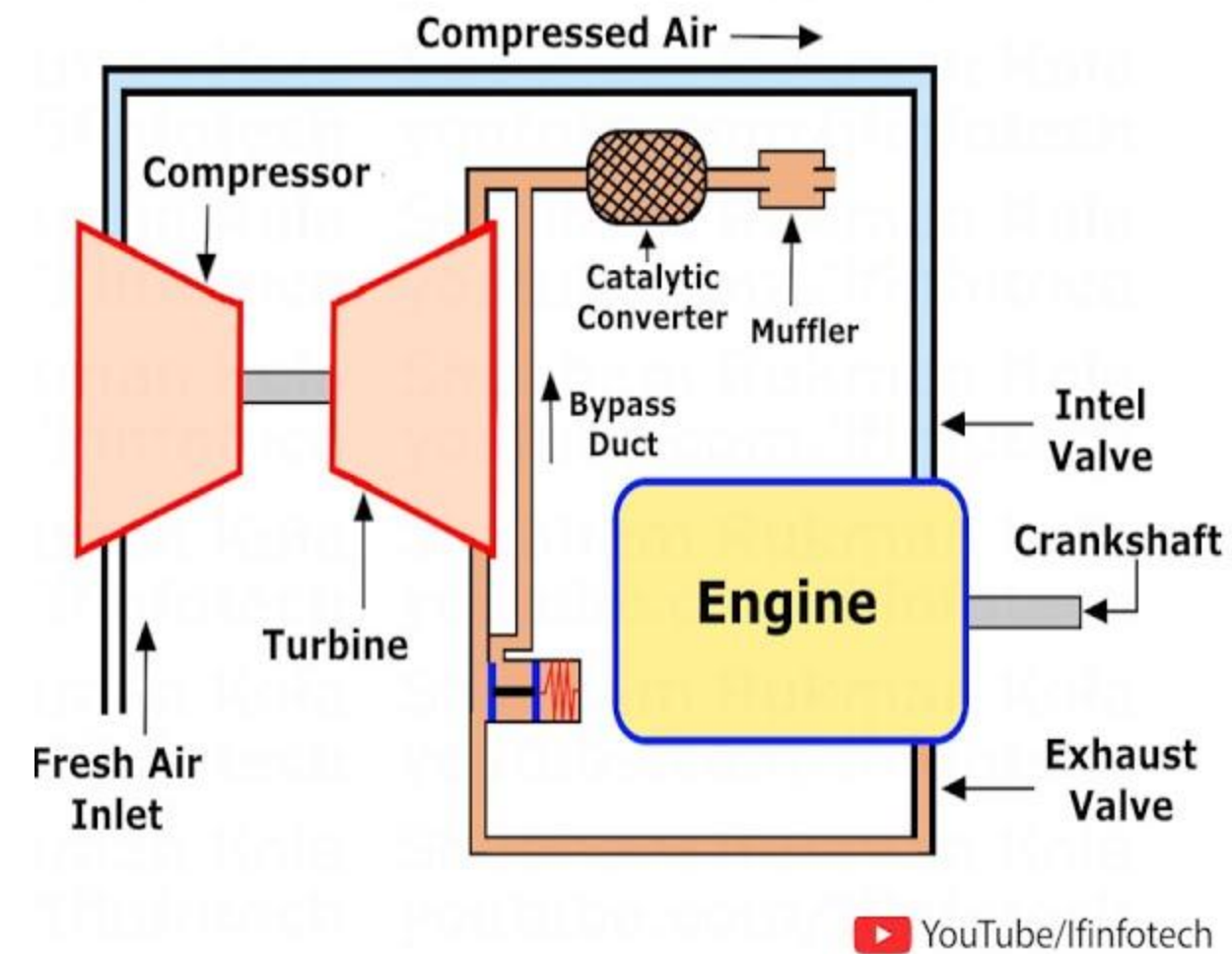
2. Turbocharger

- A **turbocharger** uses the engine's **exhaust gases** to spin a turbine, compressing the air entering the engine.

Process:

1. **Exhaust Gas Energy** – As the engine burns fuel, exhaust gases exit the combustion chamber and pass through the turbocharger's **turbine**.
2. **Turbine Rotation** – The exhaust gases spin the turbine connected by a shaft to a **compressor** on the other side.
3. **Compressed Air** – The compressor draws in fresh air, compresses it, and sends it to the intake manifold.
4. **Intercooler (Optional)** – In some setups, the compressed air passes through an **intercooler** to cool down before entering the engine, increasing efficiency.

Turbocharger





Which of the following statement is TRUE about spark-ignition engine?

- I. SI has lower compression ratio than CI
- II. SI is more challenging to start in cold conditions than CI
- III. SI typically requires more frequent maintenance than CI.

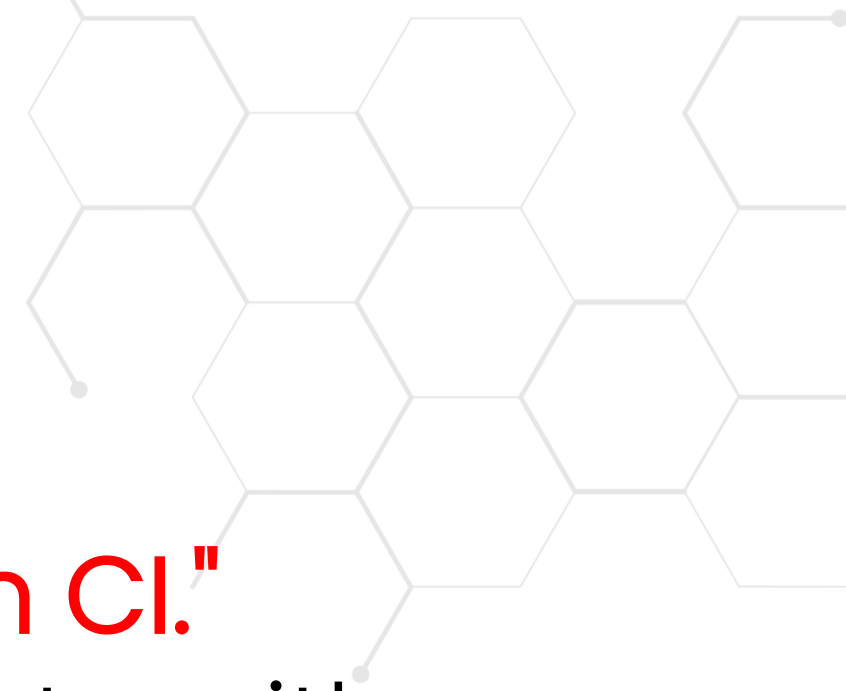
- a. I only
 - b. I, III
 - c. II, III
 - d. I, II, III
- 



Which of the following statement is TRUE about spark-ignition engine?

- I. SI has lower compression ratio than CI
- II. SI is more challenging to start in cold conditions than CI
- III. SI typically requires more frequent maintenance than CI.

- a. I only
- b. I, III**
- c. II, III
- d. I, II, III



Statement I: **"SI has lower compression ratio than CI."**

- **True:** Spark Ignition (SI) engines typically operate with a lower compression ratio compared to Compression Ignition (CI) engines. SI engines generally have compression ratios ranging from 8:1 to 12:1, while CI engines have higher compression ratios, usually between 14:1 and 25:1.



Statement II: "SI is more challenging to start in cold conditions than CI."

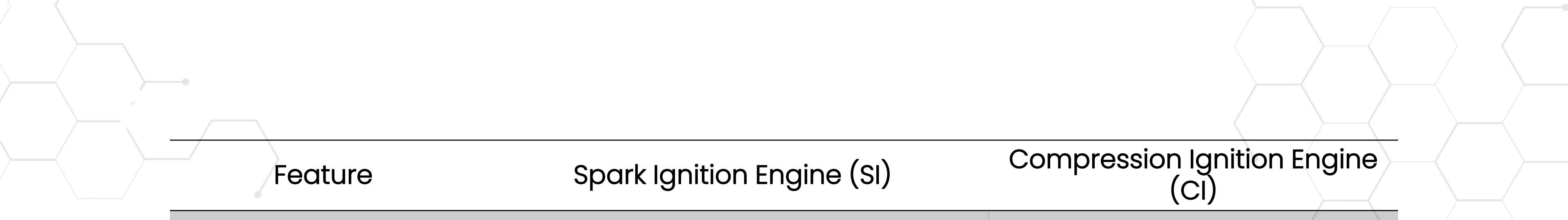
- **False:** SI engines start easier in cold conditions as they use spark plugs, while CI engines struggle due to diesel's high viscosity and compression requirements.



Statement III: "SI typically requires more frequent maintenance than CI."

- **True:** SI engines often require more frequent maintenance compared to CI engines. This is due to factors such as the need to replace spark plugs, which can wear out faster, and the nature of gasoline engines that tend to accumulate carbon deposits more quickly than diesel engines.

Feature	Spark Ignition Engine (SI)	Compression Ignition Engine (CI)
Ignition Method	Uses a spark plug to ignite the air-fuel mixture.	Relies on high compression to ignite fuel.
Fuel Type	Typically uses gasoline or lighter fuels.	Generally, uses diesel or heavier fuels.
Compression Ratio	Lower compression ratio (5:1 to 8:1)/(6:1 to 12:1)	Higher compression ratio (14:1 to 22:1)
Efficiency	Lower thermal efficiency (25-30%)	Higher thermal efficiency (30-36%)
Power Output	Produces less torque at low speeds.	Generates more torque, especially at low speeds.



Feature	Spark Ignition Engine (SI)	Compression Ignition Engine (CI)
Engine Design	Usually lighter and simpler in design.	Heavier and more robust due to high pressures.
Fuel consumption	High	Low
Maintenance	Typically requires more frequent maintenance.	More durable, but repairs can be more complex.
Starting Mechanism	Easier to start in cold conditions.	More challenging to start in cold conditions.



Types of ICE that uses diesel as fuel.

- 
- a. Spark-Ignition Engine
 - b. Compression-Ignition Engine
 - c. Multiple-cylinder engine
 - d. None of the above



Types of ICE that uses diesel as fuel.

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- a. Spark-Ignition Engine
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 - c. Multiple-cylinder engine
 - d. None of the above



Spark Ignition corresponds
to the _____ Cycle
which is a constant
_____ combustion.

- 
- a. Diesel; volume
 - b. Diesel; pressure
 - c. Otto; volume
 - d. Otto; pressure



Spark Ignition corresponds
to the _____ Cycle
which is a constant
_____ combustion.

- 
- a. Diesel; volume
 - b. Diesel; pressure
 - c. Otto; volume
 - d. Otto; pressure

- 
- 
- **Spark Ignition (SI)**
 - Otto Cycle
 - constant volume.
 - **Compression Ignition (CI)**
 - the Diesel Cycle
 - constant pressure.



A certain 4-cylinder engine has a 130.2 mm bore, a 127 mm stroke, and a clearance volume of 0.119 L in each cylinder. Is the engine a diesel or an Otto Cycle engine?

- a. Diesel
- b. Otto
- c. Neither of the two
- d. Not enough information


$$\text{Compression Ratio} = \frac{\text{Total Volume}}{CV} = \frac{PD + CV}{CV}$$

$$PD: \quad \text{Piston Displacement} = \frac{\pi D^2}{4} L$$

where L = length of stroke

CR of Diesel = 14 – 22

CR of Otto = 6 – 12

A certain 4-cylinder engine has a 130.2 mm bore, a 127 mm stroke, and a clearance volume of 0.119 L in each cylinder. Is the engine a diesel or an Otto Cycle engine?

$$PD = \frac{\pi D^2}{4} L = \frac{\pi (0.1302 \text{ m})^2}{4} (0.127 \text{ m}) = 1.6909 \times 10^{-3} \text{ m}^3$$

$$CV = 0.119 \text{ L} \times \frac{1 \text{ m}^3}{1000 \text{ L}} = 1.19 \times 10^{-4} \text{ m}^3$$

$$CR = \frac{(1.6909 \times 10^{-3} \text{ m}^3) + (1.19 \times 10^{-4} \text{ m}^3)}{1.19 \times 10^{-4} \text{ m}^3}$$

CR = 15.2 (Diesel)



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- a. Diesel
- b. Otto
- c. Neither of the two
- d. Not enough information



A V-4 engine has a total volume of 0.25 cu. m. and a clearance volume of 0.021 cu. m. in each cylinder. Is the engine a diesel or an Otto Cycle engine?

- a. Diesel
- b. Otto
- c. Neither of the two
- d. Not enough information



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$$CR = \frac{\text{Total Volume}}{CV} = \frac{0.25 \text{ cu.m}}{0.021 \text{ cu. m.}} = 11.9 \text{ (Otto)}$$



Higher octane means
the fuel can _____.

- 
- a. ignite at lower temperatures
 - b. burn more efficiently with less compression
 - c. withstand more compression before igniting
 - d. produce more power with fewer emissions



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the fuel can _____.

- 
- a. ignite at lower temperatures
 - b. burn more efficiently with less compression
 - c. withstand more compression before igniting
 - d. produce more power with fewer emissions



Cetane = Quick ignition (important for diesel).
Octane = Knocking resistance (important for gasoline).



Otto cycle gives a
_____ theoretical
efficiency than Diesel
cycle.

- 
- a. equal
 - b. lower
 - c. higher
 - d. zero



Otto cycle gives a
_____ theoretical
efficiency than Diesel
cycle.

- a. equal
 - b. lower
 - c. higher
 - d. zero
- 



Otto cycle gives a **higher** theoretical efficiency than Diesel cycle.

In actual practice, it is not higher, because the **compression ratio of Otto-cycle engine is limited by the fuel characteristics.**



During a performance test of an ICE, 80% of the maximum rated power was attained, but overheating and component failure occurred after a 7-hour continuous running test. Does the engine meet the requirements based on PAES 116?

- a. Yes
- b. No
- c. Maybe
- d. Not enough information



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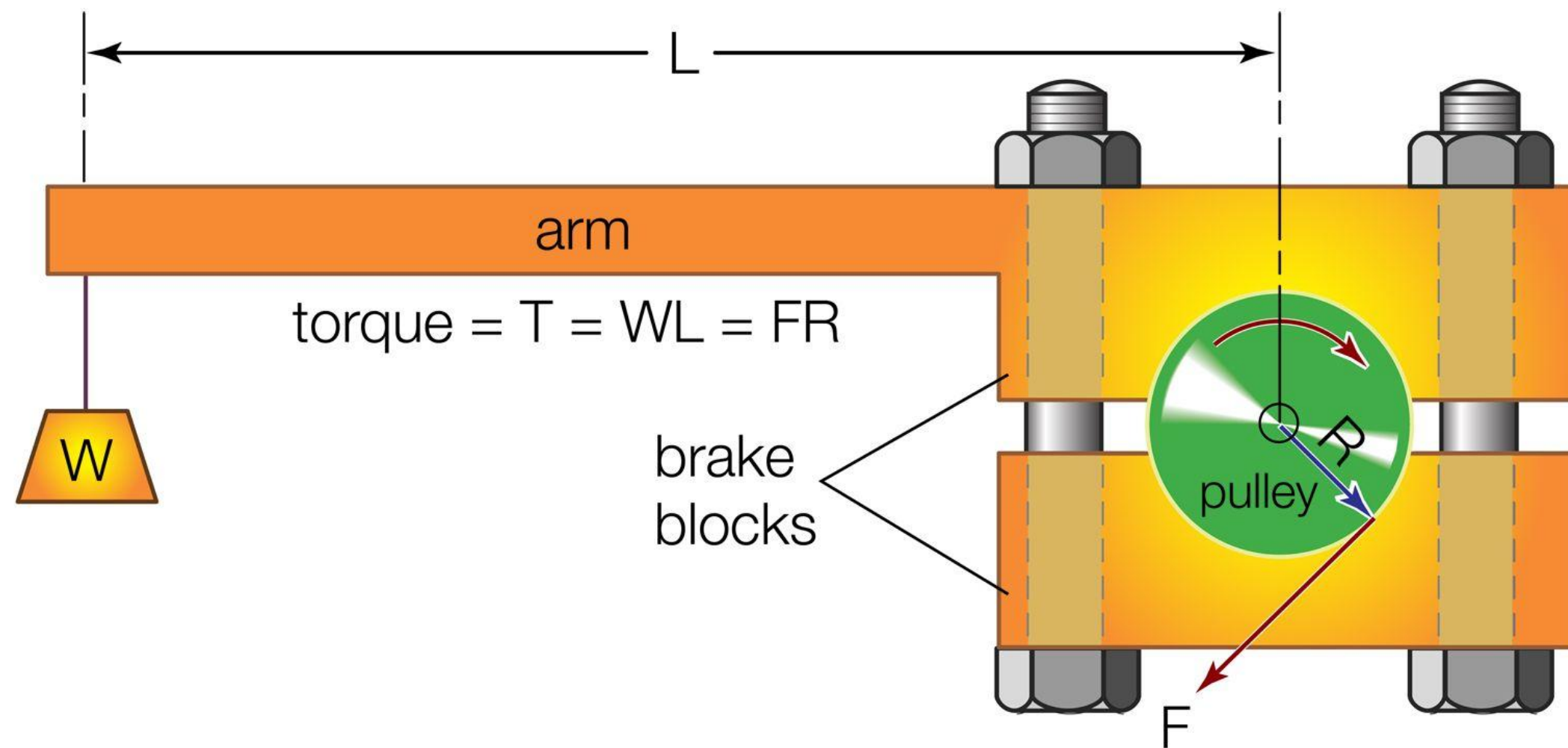
- a. Yes
- b. No
- c. Maybe
- d. Not enough information

PERFORMANCE REQUIREMENTS FOR ICE (PAES 116)

- The engine must start easily.
- It must reach **at least 80%** of the **rated maximum power** during load tests.
- The engine must achieve either the rated continuous power or **80% of the maximum power at the rated speed**.
- No breakdowns or malfunctions (e.g., overheating) should occur during a **5-hour continuous test**.
- Noise levels at 50 mm from the operator's ear must **not exceed 92 dB(A)**.

If a belt produces a tangential force of 132N on a pulley having a radius of 0.576m at a speed of 352rpm. What will be the power?

- a. 2.75 kW
- b. 2.80 kW
- c. 2.85 kW
- d. 2.90 kW



$$P = 2\pi FLN = 2\pi TN$$

P = power
 F = force
 L = length of arm
 N = rpm

$$1 \text{ HP} = 33,000 \text{ ft} - \frac{\text{lb}}{\text{min}}$$

$$1 \text{ HP} = 746 \text{ Watts}$$

$$1 \text{ HP} = 42.41 \text{ BTU/min}$$

If a belt produces a tangential force of 132N on a pulley having a radius of 0.576m at a speed of 352rpm. What will be the power?

- a. 2.75 kW
- b. 2.80 kW**
- c. 2.85 kW
- d. 2.90 kW



The engine crankshaft produces 30hp at 1700rpm. What is the torque exerted?

- a. 53.2 ft-lb
- b. 74.8 ft-lb
- c. 92.7 ft-lb
- d. NOTA



The engine crankshaft produces 30hp at 1700rpm. What is the torque exerted?

- a. 53.2 ft-lb
- b. 74.8 ft-lb
- c. 92.7 ft-lb
- d. NOTA

$$P = 2\pi FLN$$

$$P = 2\pi TN$$

$$30 \text{ hp} \left(\frac{33,000 \text{ ft-lb/min}}{1 \text{ hp}} \right) = 2\pi(T)(1700 \text{ rpm})$$

$$T = 92.68 \text{ ft-lb}$$

A decorative pattern of light gray hexagons is located at the top of the slide, spanning across the width. Some hexagons are solid, while others are outlines.

A hydro tiller is driven by a 16-HP gasoline engine that has a 10-liter capacity fuel tank fully loaded with fuel. After 6 hours of operation, it was found out that the tank only contains 6 liters of fuel. What is the fuel consumption of hydro tiller?

- a. 0.1 lph
- b. 0.67 lph
- c. 0.89 lph
- d. 1 lph


$$\text{Fuel consumption} = \frac{\Delta \text{Fuel volume}}{\# \text{ of hours of operation}}$$

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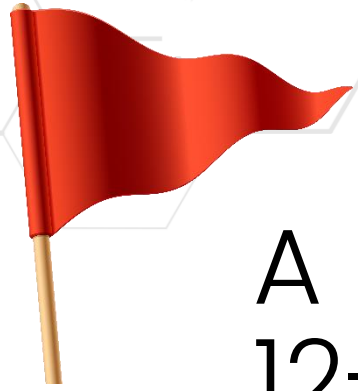
$$\text{Fuel consumption} = \frac{(10 \text{ L} - 6 \text{ L})}{6 \text{ hrs}}$$

$$\text{Fuel consumption} = 0.67 \text{ lph}$$



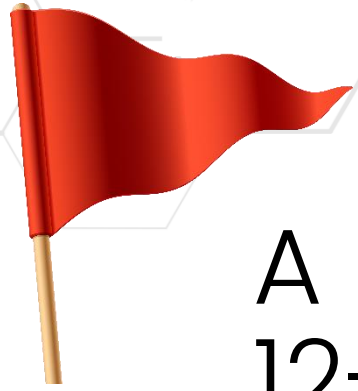
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- d. 1 lph



A hydro tiller powered by a 20-HP gasoline engine has a 12-liter fuel tank filled to capacity. The operation started at 6AM. By 11AM, only 4 liters of fuel remain. What is the hydro tiller's fuel consumption and its specific fuel consumption?

- a. 0.8; 0.24
- b. 0.8; 0.04
- c. 1.6; 0.08
- d. 1.6; 0.16



A hydro tiller powered by a 20-HP gasoline engine has a 12-liter fuel tank filled to capacity. The operation started at 6AM. By 11AM, only 4 liters of fuel remain. What is the hydro tiller's fuel consumption and its specific fuel consumption?

- a. 0.4; 0.24
- b. 0.8; 0.04
- c. 1.6; 0.08
- d. 2.4; 0.16

A hydro tiller powered by a 20-HP gasoline engine has a 12-liter fuel tank filled to capacity. The operation started at 6AM. By 11AM, only 4 liters of fuel remain. What is the hydro tiller's fuel consumption and its specific fuel consumption?

$$\text{Fuel consumption} = \frac{\Delta \text{Fuel volume}}{\# \text{ of hours of operation}}$$

$$\text{Fuel consumption} = \frac{(12 \text{ L} - 4 \text{ L})}{5 \text{ hrs}}$$

$$\text{Fuel consumption} = 1.6 \text{ lph}$$

$$\text{Specific Fuel Consumption} = \frac{\text{Fuel consumption}}{\text{Power}}$$

$$\text{Specific Fuel Consumption} = \frac{1.6 \text{ lph}}{20 \text{ hp}}$$

$$\text{Specific Fuel Consumption} = 0.08 \frac{\text{L}}{\text{hp-hr}}$$

An engine operates at a brake horsepower of 10 hp. If friction horsepower is 2.5 hp, calculate engine's mechanical efficiency.

- a. 50%
- b. 60%
- c. 70%
- d. 80%



$$\text{Mechanical Efficiency} = \frac{\text{BHP}}{\text{IHP}}$$

$$\text{IHP} = \frac{PLANn}{C} = \text{BHP} + \text{FHP}$$

$$\text{BHP} = 2\pi TN$$

An engine operates at a brake horsepower of 10 hp. If friction horsepower is 2.5 hp, calculate engine's mechanical efficiency.

$$\text{IHP} = \text{BHP} + \text{FHP}$$

$$\text{IHP} = 10 \text{ hp} + 2.5 \text{ hp}$$

$$\text{IHP} = 12.5 \text{ hp}$$

$$\text{Mechanical Efficiency} = \frac{\text{BHP}}{\text{IHP}} = \frac{10 \text{ hp}}{12.5 \text{ hp}} = 80\%$$

An engine operates at a brake horsepower of 10 hp. If friction horsepower is 2.5 hp, calculate engine's mechanical efficiency.

- a. 50%
- b. 60%
- c. 70%
- d. 80%

A 1-cylinder, 4-stroke gasoline engine, with a displacement of 97.64 in³, produces 83 lb-ft of torque while consuming 3.34 gallon/hr of fuel while running at 3100 rpm. The Indicated mean effective pressure is 164 psi. Use $HV = 20,380$ BTU/lb

23. Compute for indicated thermal.

a. 0.30 b. 0.38 c. 0.78 d. 0.82

24. Compute for brake thermal.

a. 0.30 b. 0.38 c. 0.78 d. 0.82

25. Compute for efficiency of engine.

a. 0.30 b. 0.38 c. 0.78 d. 0.82


$$\text{Indicated Thermal} = \frac{\text{IHP}}{\text{Fuel Energy}}$$

$$\text{Brake Thermal} = \frac{\text{BHP}}{\text{Fuel Energy}}$$

$$\text{Fuel Energy} = \text{Heating Value} \times \text{Fuel Consumption}$$



CONVERSION:

$$1 \text{ hp-hr} = 2545 \text{ BTU}$$

$$1 \text{ hp} = 33,000 \text{ ft-lb/min}$$

$$\text{Density of gasoline} = 6.13 \text{ lb/gal}$$


A 1-cylinder, 4-stroke gasoline engine, with a displacement of 97.64 in³, produces 83 lb-ft of torque while consuming 3.34 gallon/hr of fuel while running at 3100 rpm. The Indicated mean effective pressure is 164 psi. Use HV = 20380 BTU/lb

FOR INDICATED THERMAL:

$$\text{Fuel Energy} = \left(\frac{20380 \text{ BTU}}{\text{lb}} \times \frac{1 \text{ hp-hr}}{2,545 \text{ BTU}} \right) \times \left(\frac{3.34 \text{ gallon}}{\text{hr}} \times \frac{6.13 \text{ lb}}{\text{gallon}} \right) = 163.95 \text{ hp}$$

$$\text{IHP} = \frac{PLANn}{C} = \frac{\left(164 \frac{\text{lb}}{\text{in}^2} \times 97.64 \text{ in}^3 \times \frac{1 \text{ ft}}{12 \text{ in}} \times 3100 \text{ rpm} \times 1 \right)}{2} \times \frac{1 \text{ hp}}{33000 \text{ ft-lb/min}} = 62.677 \text{ hp}$$

$$\text{Indicated thermal} = \frac{\text{IHP}}{\text{Fuel Power}} = \frac{62.677 \text{ hp}}{163.95 \text{ hp}}$$

Indicated thermal = 0.38

A 1-cylinder, 4-stroke gasoline engine, with a displacement of 97.64 in³, produces 83 lb-ft of torque while consuming 3.34 gallon/hr of fuel while running at 3100 rpm. The Indicated mean effective pressure is 164 psi. Use HV = 20,380 BTU/lb

FOR BRAKE THERMAL:

$$\text{BHP} = (2\pi)(83 \text{ lb-ft})(310 \text{ rpm}) \times \frac{1 \text{ hp}}{33000 \text{ ft-lb/min}} = 48.99 \text{ hp}$$

$$\text{Brake thermal} = \frac{\text{BHP}}{\text{Fuel Power}} = \frac{48.99 \text{ hp}}{163.95 \text{ hp}}$$

$$\text{Brake thermal} = 0.30$$

A 1-cylinder, 4-stroke gasoline engine, with a displacement of 97.64 in³, produces 83 lb-ft of torque while consuming 3.34 gallon/hr of fuel while running at 3100 rpm. The Indicated mean effective pressure is 164 psi. Use HV = 20,380 BTU/lb

FOR MECHANICAL EFFICIENCY OF ENGINE:

$$ME = \frac{BHP}{IHP} = \frac{48.99 \text{ hp}}{62.88 \text{ hp}}$$

$$ME = 0.78 \text{ or } 78\%$$

A 1-cylinder, 4-stroke gasoline engine, with a displacement of 97.64 in³, produces 83 lb-ft of torque while consuming 3.34 gallon/hr of fuel while running at 3100 rpm. The Indicated mean effective pressure is 164 psi. Use HV = 20,380 BTU/lb

23. Compute for indicated thermal.

a. 0.30 **b. 0.38** c. 0.78 d. 0.82

24. Compute for brake thermal.

a. 0.30 b. 0.38 c. 0.78 d. 0.82

25. Compute for efficiency of engine.

a. 0.30 b. 0.38 **c. 0.78** d. 0.82

A farmer is preparing to plow a large field with heavy, clay soil using a disc plow. Later, they will switch to a rotary tiller to prepare the seedbed. However, the tractor struggles during plowing but performs well during the tillage with the rotary tiller.

Which type of power—drawbar power or PTO power—is most critical for each operation?

- a. Drawbar power for both plowing and seedbed preparation
- b. PTO power for both plowing and seedbed preparation
- c. Drawbar power for plowing; PTO power for seedbed preparation
- d. PTO power for plowing; drawbar power for seedbed preparation

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- d. PTO power for plowing; drawbar power for seedbed preparation

Drawbar Power: refers to the tractor's ability to pull heavy loads, such as plows, harrows, or trailers.

Usage: It's essential for operations that require traction, like plowing, tilling, and hauling.

PTO Power: PTO power refers to the power the engine delivers to external implements via the PTO shaft (a rotating shaft at the rear or side of the tractor).

Usage: It's critical for operating equipment with rotating parts, such as rotary tillers, and mowers

Hydraulic Power: Some tractors also provide hydraulic power for lifting or operating attachments (like loaders or backhoes) using hydraulic cylinders.



Crank mechanism is used to convert reciprocating motion to rotary motion and vice versa. It consists of following components EXCEPT _____.

- 
- a. Crankcase
 - b. Crankshaft
 - c. Vibration dumper
 - d. Valves



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- a. Crankcase
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 - d. Valves

A farmer is troubleshooting their petrol-powered tractor, which uses a carburetor for the air-fuel mixture. They notice that the engine sputters and produces less power than usual, especially when accelerating. After inspecting the carburetor, they realize that one of its functions isn't being performed properly.

Which of the following carburetor functions, if not working correctly, is most likely the cause of the engine's poor performance?

- a. Regulating the airflow in the engine
- b. Supplying the required amount of fuel to maintain the fuel/air mixture
- c. Providing constant engine lubrication
- d. Ensuring the exhaust gases are fully cleaned by the exhaust system

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A. Regulating the airflow in the engine

Regulating airflow is a key function of the carburetor, but insufficient fuel supply would have a more direct impact on the engine sputtering and losing power.



B. Supplying the required amount of fuel to maintain the fuel/air mixture

This is the correct answer. If the carburetor is not supplying enough fuel, the air-fuel ratio becomes too lean, causing poor engine performance and sputtering, especially under acceleration.



C. Providing constant engine lubrication

Lubrication is handled by the engine oil system, not the carburetor.



D. Ensuring the exhaust gases are fully cleaned by the exhaust system

The exhaust system ensures emissions are controlled but **doesn't affect how the carburetor performs or how fuel is mixed with air.**

A farmer is operating a diesel-powered tractor during the cold season and notices several issues: the engine is difficult to start, and the tractor emits excessive smoke and carbon deposits are forming in the exhaust system. Upon inspection, the farmer suspects that the quality and properties of the fuel may be the cause.

Which of the following fuel properties is most likely causing the difficulties in starting the engine and the increase in smoke and carbon deposits?

- a. Heat value (Calorific value)
- b. Volatility
- c. Flash point
- d. Pour point

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FUEL PROPERTIES

1. Heat Value (Calorific Value):

- Measures the energy released when fuel burns (in J/kg).
- Higher impurities reduce heat value.
- Measured with a **calorimeter**.

2. Specific Gravity:

- Ratio of fuel's density to water's density.
- Affects fuel atomization and spray in the engine.
- Heavier fuels usually have more energy.
- Measured with a **hydrometer**.

3. Volatility:

- How easily fuel turns to vapor at a given temperature.
- Low volatility = more carbon deposits and smoke.
- In diesel, measured by the **90% distillation temperature**.



4. Flash Point:

- Lowest temperature where fuel gives off vapors that ignite.
- A low flash point = higher fire hazard.

5. Fire Point:

- Temperature at which vapors keep burning after ignition.
- Usually 100–210°C higher than the flash point.

6. Pour Point:

- Temperature where fuel stops flowing.
- In cold weather, a high pour point can clog filters and reduce spray quality.

“A farmer is operating a diesel-powered tractor during the cold season and notices several issues: the engine is difficult to start, and the tractor emits excessive smoke and carbon deposits are forming in the exhaust system”

- ✗ While calorific value affects the energy produced by the fuel, it doesn't directly cause starting problems or smoke.
- ✓ Low volatility means the fuel does not vaporize efficiently, leading to incomplete combustion, more carbon deposits, and smoke. This makes cold starts difficult because the fuel does not burn properly.
- ✗ Flash point indicates the temperature at which vapors ignite. While important for safety, it isn't directly responsible for excessive smoke or cold starting issues.
- ✗ Pour point affects fuel flow at low temperatures. While it could make starting harder if the fuel solidifies, the issue described points more toward poor combustion, which relates to volatility.



A farmer's engine is prone to knocking, especially on hot days when the engine runs for long hours. To avoid engine damage and improve combustion, the farmer asks for ways to prevent detonation.

Which of the following would be the best solution to prevent engine detonation in this situation?

- a. Increasing the valve timing
- b. Using a higher-octane fuel with anti-knock agents
- c. Running the engine at low speeds constantly
- d. Increasing the air-fuel ratio (more air, less fuel)



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ASPECT	RIGHT IGNITION	WRONG IGNITION (DETONATION)
Timing	Ignition occurs at the optimal moment when the piston is near the top of its stroke.	Ignition happens too early (pre-ignition) or too violently, causing an explosion.
Burning rate	Fuel burns smoothly and steadily, allowing for gradual pressure buildup.	Fuel burns too quickly, resulting in a sudden pressure spike.
Pressure	Pressure builds in a controlled manner, optimizing power output.	Excessive pressure builds up too soon, leading to knocking sounds.
Sound	Produces a smooth and consistent sound during operation.	Produces a loud "knocking" or "pinging" sound.



Effects of detonation

1. Inefficient combustion.
2. Loss power.
3. Local overheating.
4. Mechanical engine failure.



Prevention of Detonation

1. Anti-knock agents.
2. Cooling of the charge.
3. Reducing the time factor.

A farmer notices that their tractor engine runs too fast when it's under heavy load, causing it to vibrate and become difficult to control. The farmer wonders if there is a way to maintain a steady speed regardless of the load.

What component should the farmer check to help regulate the engine speed automatically when the load increases?

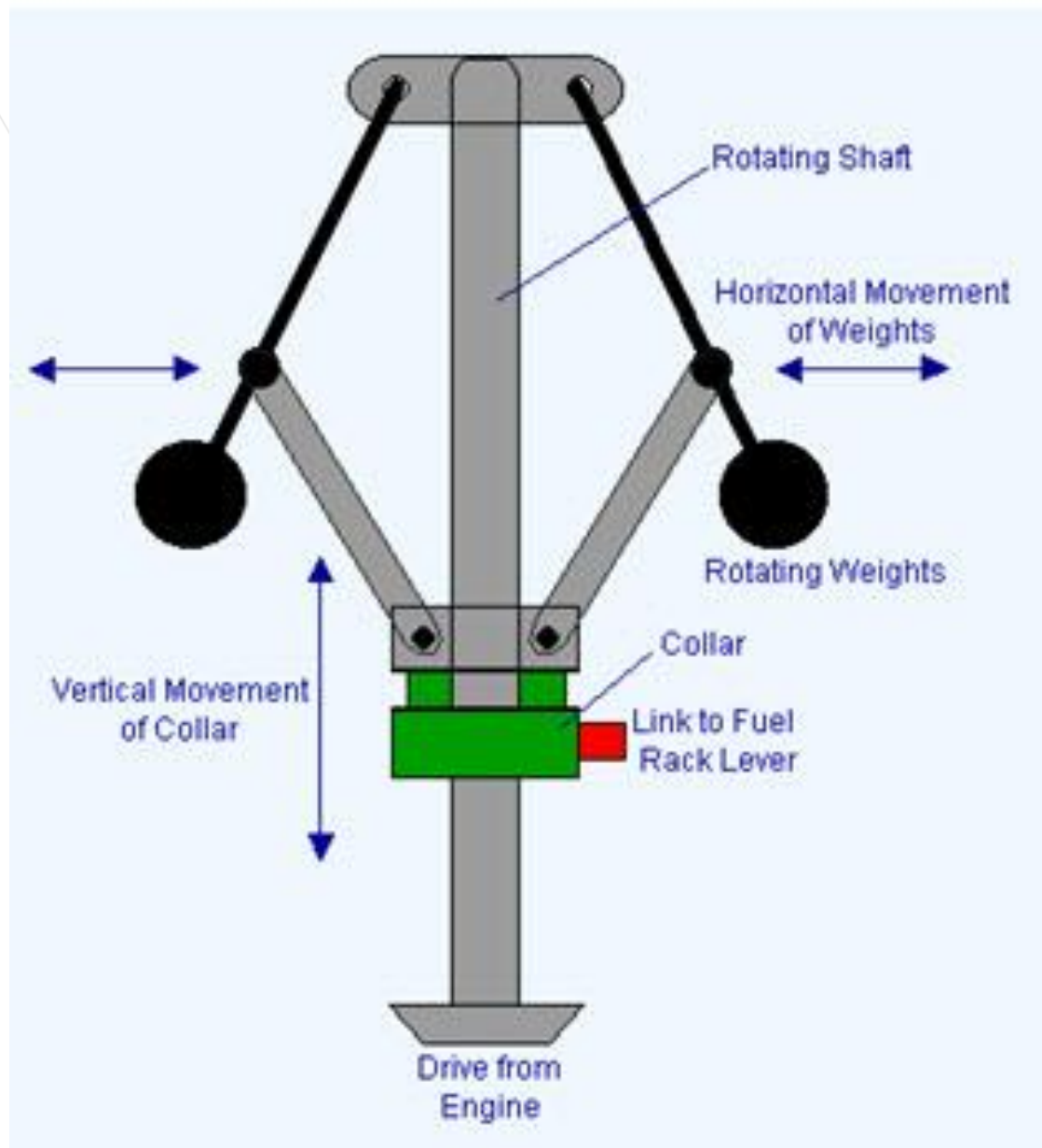
- a. Carburetor
- b. Oil filter
- c. Engine governor
- d. Fuel tank

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What component should the farmer check to help regulate the engine speed automatically when the load increases?

- a. Carburetor
- b. Oil filter
- c. Engine governor
- d. Fuel tank

A governor is a device that controls engine speed by adjusting fuel delivery.



- ❑ **Balance of Forces:** It uses a spring and flyweights, where centrifugal force from the flyweights balances the spring force at a specific speed.
- ❑ **Speed Increase:** If engine speed increases, centrifugal force rises, leading the governor to reduce fuel delivery and slow the engine down.
- ❑ **Speed Decrease:** If engine speed drops, the governor increases fuel delivery to restore the desired speed.
- ❑ **Automatic Control:** The governor automatically manages all engine speeds, including idling.
- ❑ **Use:** Governors are commonly integrated into fuel injection pumps in diesel engines.



If a cow can provide a draft of 60 kg, how much draft can be provided by 2 cow together?

- a. 120 kg
 - b. 18 kg
 - c. 114 kg
 - d. 102 kg
- 



If a cow can provide a draft of 60 kg, how much draft can be provided by 2 cow together?

- a. 120 kg
 - b. 18 kg
 - c. 114 kg
 - d. 102 kg
- 



Harnessing Factor of Draft Animals



No. of Animals	Factor
1	1.0
2	1.9
3	2.5
4	3.1
5	3.5
6	3.8

If a cow can provide a draft of 60 kg, how much draft can be provided by 2 cow together?

Draft = No. of draft animals x draft x Harnessing Factor

Draft = 2 x 60 kg x 0.95

Draft = 114 kg



A person is working for 4 hours in a farm harvesting. Estimate the power that can be generated by the person doing the job.

- a. 0.4 hp
 - b. 0.21 hp
 - c. 0.13 hp
 - d. 0.1 hp
- 



A person is working for 4 hours in a farm harvesting. Estimate the power that can be generated by the person doing the job.

- a. 0.4 hp
 - b. 0.21 hp
 - c. 0.13 hp
 - d. 0.1 hp
- 



Human Power:

$$P = 0.35 - 0.092[\log(t)]$$

t is expressed in minutes

Source: Agricultural Power And Energy Sources By Engr. Alexis T. Belonio

A person is working for 4 hours in a farm harvesting. Estimate the power that can be generated by the person doing the job.

$$P = 0.35 - 0.092[\log(t)]$$

$$P = 0.35 - 0.092\left[\log\left(4 \text{ hr} \times 60 \frac{\text{min}}{\text{hr}}\right)\right]$$

$$P = 0.1310 \text{ HP}$$

Power Output

1. Human can develop a total power of 0.5 hp from the food he eats.
2. 0.1 Hp is available for useful work and the remaining is use for body functions.
3. Man at 20 years of age can generate 15% more power and less by 20% at 60 years of age.

Energy Requirement and Efficiency

1. Minimum energy requirement is 2000 cal per day.
2. Conversion efficiency is 25%.
3. Working efficiency decreases under hot and humid climate and effective to work under normal condition.



What is the maximum
load a carabao can
pull?

- a. 250 kg
 - b. 300 kg
 - c. 350 kg
 - d. 400 kg
- 



What is the maximum
load a carabao can
pull?

- a. 250 kg
 - b. 300 kg
 - c. 350 kg
 - d. 400 kg
- 



According to *Test data of the National Carabao Research Center, Institute of Animal Science, UPLB, 1990:*

“The test draft animals could not pull the load of 300 kg and beyond.”

A 4-cylinder, 100 mm × 100 mm [4 in × 4 in] engine turns 1600 rpm.

35. What is the total engine displacement?

a. 0.00314 m³ b. 0.00785 m³ c. 0.00912 m³ d. 0.00322 m³

36. If the instantaneous piston head pressure is 690 kPa [100 psi] when the shaft has rotated 90° after top dead center position, what is the torque on the crankshaft?

a. 290.75 N-m b. 270.95 N-m c. 520.79 N-m d. 207.59 N-m

37. If the mean effective pressure per power stroke is 550 kPa [80 psi], what is the indicated power of the engine?

a. 30.88 hp b. 1123.5 W c. 3.9 hp d. 12.60 kW

38. If the individual cylinder's clearance volume is 130 cm³ [8 in³], what is the engine's compression ratio?

a. 8.1:1 b. 9.02:1 c. 10:1 d. None of the above


$$\text{Total Displacement} = (\text{Area of Piston} \times \text{Length}) \times \# \text{Cylinder}$$

Torque = Force x Moment Arm
Where: Force = Area of Piston x Pressure
Moment arm = $\frac{1}{2}$ of bore size

$$\text{IHP} = \frac{PLANn}{C}$$

$$\text{Compression Ratio} = \frac{PD + CV}{CV}$$

A 4-cylinder, 100 mm × 100 mm [4 in × 4 in] engine turns 1600 rpm.

35. What is the total engine displacement?

$$A = \frac{\pi(D)^2}{4} = \frac{\pi(0.10 \text{ m})^2}{4} = \frac{1}{400} \pi \text{ m}^2$$

$$\text{PD of each cylinder} = \frac{1}{400} \pi \text{ m}^2 \times 0.10 \text{ m} = 7.85 \times 10^{-4} \text{ m}^3$$

$$\text{Total Displacement} = (A \times L) \times \# \text{ of cylinder}$$

$$\text{Total Displacement} = (7.85 \times 10^{-4}) \times 4$$

$$\text{Total Displacement} = 3.14 \times 10^{-3} \text{ m}^3$$

A 4-cylinder, 100 mm × 100 mm [4 in × 4 in] engine turns 1600 rpm.

36. If the instantaneous piston head pressure is 690 kPa [100 psi] when the shaft has rotated 90° after top dead center position, what is the torque on the crankshaft?

Force = Area of Piston x Pressure

$$\text{Force} = \left(\frac{1}{400} \pi \text{ m}^2\right) \text{m}^2 \times 690 \text{ kN/m}^2$$

$$\text{Force} = 5.419 \text{ kN} = 5419 \text{ N}$$

Note: 1kPa = 1kN/m²

$$\text{Moment arm} = \frac{1}{2} \text{ of bore size} = \frac{1}{2} (0.10 \text{ m}) = 0.05 \text{ m}$$

$$\text{Torque} = 5419 \text{ N} \times 0.05\text{m} = \mathbf{270.95 \text{ N-m}}$$

A 4-cylinder, 100 mm × 100 mm [4 in × 4 in] engine turns 1600 rpm.

37. If the mean effective pressure per power stroke is 550 kPa [80 psi], what is the indicated power of the engine.

$$\text{IHP} = \frac{PLANn}{C} = \frac{(550 \frac{\text{kN}}{\text{m}^2})(0.10 \text{ m})(\frac{1}{400}\pi \text{ m}^2)(1600 \frac{\text{rev}}{\text{min}})(\frac{1 \text{ min}}{60 \text{ sec}})(4)}{2} = 23.038 \text{ kW}$$

$$\text{IHP} = 23.038 \text{ kW} \times \frac{1 \text{ hp}}{0.746 \text{ kW}} = \mathbf{30.88 \text{ hp}}$$

A 4-cylinder, 100 mm × 100 mm [4 in × 4 in] engine turns 1600 rpm.

38. If the individual cylinder's clearance volume is 130 cm³ [8 in³], what is the engine's compression ratio?

$$CV = 130 \text{ cm}^3 \left(\frac{1 \text{ m}}{100 \text{ cm}} \right)^3 = 1.30 \times 10^{-4}$$

$$CR = \frac{PD}{CV} = \frac{(7.85 \times 10^{-4} \text{ m}^3 + 1.30 \times 10^{-4} \text{ m}^3)}{1.30 \times 10^{-4} \text{ m}^3}$$

$$CR = \mathbf{7.04:1}$$

A 4-cylinder, 100 mm × 100 mm [4 in × 4 in] engine turns 1600 rpm.

35. What is the total engine displacement?

a. 0.00314 m³ b. 0.00785 m³ c. 0.00912 m³ b. 0.00322 m³

36. If the instantaneous piston head pressure is 690 kPa [100 psi] when the shaft has rotated 90° after top dead center position, what is the torque on the crankshaft?

a. 290.75 N-m **b. 270.95 N-m** c. 520.79 N-m b. 207.59 N-m

37. If the mean effective pressure per power stroke is 550 kPa [80 psi], what is the indicated power of the engine?

a. 30.88 hp b. 1123.5 W a. 3.9 hp b. 12.60 kW

38. If the individual cylinder's clearance volume is 130 cm³ [8 in³], what is the engine's compression ratio?

a. 8.1:1 b. 9.02:1 c. 10:1 **d. None of the above**



A 6-cylinder, 200 mm × 150 mm engine turns 3150 rpm.

39. What is the total engine displacement?

a. 0.002826 m³ b. 0.2826 m³ c. 0.00283 m³ d. 0.02826 m³

40. If the instantaneous piston head pressure is 690 kPa [100 psi] when the shaft has rotated 90° after top dead center position, what is the torque on the crankshaft?

a. 1.5 kN-m b. 1.626 kN-m c. 1.5 N-m d. 1.5626 N-m

41. If the mean effective pressure per power stroke is 550 kPa [80 psi], what is the indicated power of the engine?

a. 198.34 hp b. 201 hp c. 338 hp d. 547.2 hp

42. If the individual cylinder's clearance volume is 130 cm³ [8 in³], what is the engine's compression ratio?

a. 7.56:1 b. 15.32:1 c. 37.23 :1 d. None of the above

A 6-cylinder, 200 mm × 150 mm engine turns 3150 rpm.

39. What is the total engine displacement?

$$A = \frac{\pi(D)^2}{4} = \frac{\pi(0.20 \text{ m})^2}{4} = \frac{1}{100} \pi \text{ m}^2$$

$$\text{PD of each cylinder} = \frac{1}{100} \pi \text{ m}^2 \times 0.15 \text{ m} = 4.71 \times 10^{-3} \text{ m}^3$$

$$\text{Total Displacement} = (A \times L) \times \# \text{ of cylinder}$$

$$\text{Total Displacement} = (4.71 \times 10^{-3} \text{ m}^3) \times 6$$

$$\text{Total Displacement} = 0.02826 \text{ m}^3$$

A 6-cylinder, 200 mm × 150 mm engine turns 3150 rpm.

40. If the instantaneous piston head pressure is 690 kPa [100 psi] when the shaft has rotated 90° after top dead center position, what is the torque on the crankshaft?

Force = Area of Piston x Pressure

$$\text{Force} = \left(\frac{1}{100} \pi \text{ m}^2\right) \text{m}^2 \times 690 \text{ kN/m}^2$$

$$\text{Force} = \frac{69}{10} \pi \text{ kN} = 21.677 \text{ kN}$$

Note: 1kPa = 1kN/m²

$$\text{Moment arm} = \frac{1}{2} \text{ of bore size} = \frac{1}{2} (0.15 \text{ m}) = 0.075 \text{ m}$$

$$\text{Torque} = 21.677 \text{ kN} \times 0.075 \text{ m} = \mathbf{1.626 \text{ kN-m}}$$

A 6-cylinder, 200 mm × 150 mm engine turns 3150 rpm.

41. If the mean effective pressure per power stroke is 550 kPa [80 psi], what is the indicated power of the engine.

$$\text{IHP} = \frac{PLANn}{C} = \frac{(550 \frac{\text{kN}}{\text{m}^2})(0.15 \text{ m})(\frac{1}{100}\pi \text{ m}^2)(3150 \frac{\text{rev}}{\text{min}})(\frac{1 \text{ min}}{60 \text{ sec}})(6)}{2} = \frac{2079}{16}\pi \text{ kW}$$

$$\text{IHP} = \frac{2079}{16}\pi \text{ kW} \times \frac{1 \text{ hp}}{0.746 \text{ kW}} = \mathbf{547.2 \text{ hp}}$$

A 6-cylinder, 200 mm × 150 mm engine turns 3150 rpm.

42. If the individual cylinder's clearance volume is 130 cm³ [8 in³], what is the engine's compression ratio?

$$CV = 130 \text{ cm}^3 \left(\frac{1 \text{ m}}{100 \text{ cm}} \right)^3 = 1.30 \times 10^{-4}$$

$$CR = \frac{PD}{CV} = \frac{(4.71 \times 10^{-3} \text{ m}^3 + 1.30 \times 10^{-4} \text{ m}^3)}{1.30 \times 10^{-4} \text{ m}^3}$$

$$CR = \mathbf{37.23 : 1}$$

A 6-cylinder, 200 mm × 150 mm engine turns 3150 rpm.

39. What is the total engine displacement?

a. 0.002826 m³ b. 0.2826 m³ c. 0.00283 m³ d. **0.02826 m³**

40. If the instantaneous piston head pressure is 690 kPa [100 psi] when the shaft has rotated 90° after top dead center position, what is the torque on the crankshaft?

a. 1.5 kN-m b. **1.626 kN-m** c. 1.5 N-m d. 1.5626 N-m

41. If the mean effective pressure per power stroke is 550 kPa [80 psi], what is the indicated power of the engine?

a. 198.34 hp b. 201 hp c. 338 hp d. **547.2 hp**

42. If the individual cylinder's clearance volume is 130 cm³ [8 in³], what is the engine's compression ratio?

a. 7.56:1 b. 15.32:1 c. **37.23 :1** d. None of the above



Yanmar-Yeh Hsing Diesel
Engine is made from?

- 
- a. Republic of China
 - b. Japan
 - c. Taipei
 - d. Italy



Yanmar-Yeh Hsing Diesel
Engine is made from?

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- a. Republic of China
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What engine is made in
Osaka, Japan?

- a. Honda
 - b. Kohler
 - c. Kubota
 - d. Mitsubishi Katsura
- 



What engine is made in
Osaka, Japan?

- a. Honda
 - b. Kohler
 - c. **Kubota**
 - d. Mitsubishi Katsura
- 

- 
- 
- ACME and Ducati Air Cooled Diesel Engines – Italy
 - Honda and Robin Engines – Tokyo, Japan
 - Kubota Engines – Osaka, Japan
 - Kohler Engines – USA
 - MAG Kerosene Engine – Geneva, Switzerland
 - Yanmar-Yeh Hsing Diesel Engine – Taipei



Which engine model
from the list below
provides the highest
power output?

- 
- a. ACME
 - b. Ducati
 - c. Honda
 - d. Robin



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Engine Specifications

Brand	Model	Type	Bore x Stroke	Displacement	Output Power	Maximum Torque
ACME	AON 48W	4-stroke vertical air-cooled diesel	85 mm x 85 mm	482 cc	11.3 hp / 3000 rpm	2.52 kg-m / 2058 rpm
Ducati	IS 7	4-cycle vertical air-cooled	75 mm x 78 mm	345 cc	7.5 hp / 3200 rpm	1.1 kg-m / 2500 rpm
Honda	GX 160	4-cycle air-cooled gasoline horizontal shaft	68 mm x 45 mm	163 cc	5.3 hp / 3600 rpm	0.68 kg-m / 2800 rpm
Robin	EY 15 D	Air-cooled, 4-cycle gasoline, horizontal PTO shaft	63 mm x 46 mm	143 cc	3.5 hp / 4000 rpm	-



A tractor pulls 2500 kg of load at a speed of 5kph on a flat ground. If the coefficient of rolling resistance is 0.25, what is the drawbar power that the tractor generates?

- 
- a. 15 hp
 - b. 8.51 hp
 - c. 11.42 hp
 - d. 7 hp

A tractor pulls 2500 kg of load at a speed of 5kph on a flat ground. If the coefficient of rolling resistance is 0.25, what is the drawbar power that the tractor generates?

$$\text{Rolling Resistance Force (F)} = 2500 \text{ kg} \left(9.81 \frac{\text{m}}{\text{s}^2}\right) \times 0.25$$

$$F = 6131.25 \text{ N}$$

$$\text{DBP} = F \times \text{Speed}$$

$$\text{DBP} = 6131.25 \text{ N} \times 5 \frac{\text{km}}{\text{hr}} \left(\frac{1 \text{ hr}}{3600 \text{ sec}}\right) \left(\frac{1000 \text{ m}}{1 \text{ km}}\right)$$

$$\text{DBP} = 8515.625 \text{ Watts} \times \frac{1 \text{ hp}}{746 \text{ Watts}} = \mathbf{11.42 \text{ hp}}$$



A tractor pulls 2500 kg of load at a speed of 5kph on a flat ground. If the coefficient of rolling resistance is 0.25, what is the drawbar power that the tractor generates?

- a. 15 hp
- b. 8.51 hp
- c. 11.42 hp
- d. 7 hp



The fuel burns a mixture of 58 L of isooctane and 5 L of n-heptane by volume. What is the octane rating?

- a. 90
- b. 91
- c. 92
- d. 93


$$\text{Octane Rating} = \frac{\text{Volume of Isooctane}}{\text{Total Volume}}$$

The fuel burns a mixture of 58 L of isooctane and 5 L of n-heptane by volume. What is the octane rating?

$$\text{Octane Rating} = \frac{\text{Volume of Isooctane}}{\text{Total Volume}}$$

$$\text{Octane Rating} = \frac{58 \text{ L}}{58 \text{ L} + 5 \text{ L}} = 0.92 \approx \mathbf{92\%}$$



The fuel burns a mixture of 58 L of isooctane and 5 L of n-heptane by volume. What is the octane rating?

- a. 90
- b. 91
- c. 92
- d. 93



What is the cetane rating of a mixture of 7L of N-cetane and 8L of heptamethylnonane (HMN)?

- a. 21.90
 - b. 38.02
 - c. 44.77
 - d. 54.67
- 



$$\text{Cetane Rating} = (\% \text{ N-cetane}) + 0.15(\% \text{ HMN})$$

What is the cetane rating of a mixture of 7L of N-cetane and 8L of heptamethylnonane (HMN)?

$$\text{Cetane Rating} = (\% \text{ N-cetane}) + 0.15(\% \text{ HMN})$$

$$\% \text{ N-cetane} = \frac{7 \text{ L}}{7 \text{ L} + 8 \text{ L}} = 0.4667$$

$$\% \text{ HMN} = \frac{8 \text{ L}}{7 \text{ L} + 8 \text{ L}} = 0.5333$$

$$\text{Cetane Rating} = (0.4667) + 0.15(5333)$$

$$\text{Cetane Rating} = 0.546695 \text{ or } \mathbf{54.67\%}$$



What is the cetane rating of a mixture of 7L of N-cetane and 8L of heptamethylnonane (HMN)?

- 
- a. 21.90
 - b. 38.02
 - c. 44.77
 - d. 54.67



The fuel burns a mixture of 23 L of isooctane and 7 L of N-heptane by volume. What is the octane rating?

- a. 56.67 b. 66.677 c. 77.67 d. None of the above



What is the cetane rating of a mixture of 5L of N-cetane and 5L of heptamethylnonane (HMN)?

- a. 57.5 b. 58.5 c. 59.5 d. None of the above



The fuel burns a mixture of 23 L of isooctane and 7 L of N-heptane by volume. What is the octane rating?

- a. 56.67 b. 66.677 c. 77.67 d. None of the above



What is the cetane rating of a mixture of 5L of N-cetane and 5L of heptamethylnonane (HMN)?

- a. 57.5 b. 58.5 c. 59.5 d. None of the above



What is thermal efficiency of gasoline engine which uses 0.315 kg of fuel per kW-hr?

- a. 11.6%
 - b. 24.3%
 - c. 31.5%
 - d. 40.2%
- 



Energy content of fuels



FUEL	ENERGY CONTENT	DENSITY
Regular Gasoline	47.06 MJ/kg	0.74 kg/L
Premium Gasoline	47.15 MJ/kg	0.73 kg/L
Diesel	45.46 MJ/kg	0.85 kg/L


$$\text{Thermal efficiency} = \frac{\text{Useful energy output}}{\text{Energy provided by fuel}}$$

What is thermal efficiency of gasoline engine which uses 0.315 kg of fuel per kW-hr?

$$\text{Thermal efficiency} = \frac{\text{Useful energy output}}{\text{Energy provided by fuel}}$$

$$\text{Energy from fuel} = 0.315 \text{ kg} \times 47.06 \text{ MJ/kg} = 14.82 \text{ MJ}$$

Useful energy output is stated “per kW-hr”. Thus,
 $\text{Energy output} = 1 \text{ kW-hour} = 3.6 \text{ MJ}$

$$\text{Thermal efficiency} = \frac{3.6 \text{ MJ}}{14.82 \text{ MJ}} = 0.243 = 24.3\%$$



What is thermal efficiency of gasoline engine which uses 0.315 kg of fuel per kW-hr?

- a. 11.6%
 - b. 24.3%**
 - c. 31.5%
 - d. 40.2%
- 



This value refers to the amount of solar energy per unit area received at the top of the Earth's atmosphere when the Earth is at its average distance from the Sun.

- 
- a. Solar radiation
 - b. Solar constant
 - c. Beam radiation
 - d. Solar insolation



This value refers to the amount of solar energy per unit area received at the top of the Earth's atmosphere when the Earth is at its average distance from the Sun.

- 
- a. Solar radiation
 - b. Solar constant**
 - c. Beam radiation
 - d. Solar insolation



Solar Constant is the fixed amount of solar energy that reaches the top of Earth's atmosphere, measured per second on 1 square meter facing the Sun. Think of it as the Sun's power output at Earth's distance. (1353 W/m²)

Solar Insolation is the total solar energy that hits a certain area (like the ground or a solar panel) over a period of time. It's measured in kWh/m² per day. Insolation changes depending on weather, location, and time of year.



Refers to the angle with which a ray of sunlight strikes the surface of the PV module, measured with respect to a line perpendicular to said surface.

- 
- a. Angle of inclination
 - b. Angle of reflectance
 - c. Ray of incidence
 - d. Ray of inclination



Refers to the angle with which a ray of sunlight strikes the surface of the PV module, measured with respect to a line perpendicular to said surface.

- 
- a. Angle of inclination
 - b. Angle of reflectance
 - c. Ray of incidence
 - d. Ray of inclination



Ray of incidence – the angle with which a ray of sunlight strikes the surface of the PV module, measured with respect to a line perpendicular to said surface.

Angle of inclination – the tilt of the PV module with respect to the horizontal plane

Source: PNS/BAFS 324:2021



Refers to the assembly of photovoltaic cells mounted in a frame that uses sunlight as a source of energy to generate a DC electricity

- 
- a. PV Module
 - b. Solar panel
 - c. Photovoltaic array
 - d. Two of the choices above



Refers to the assembly of photovoltaic cells mounted in a frame that uses sunlight as a source of energy to generate a DC electricity

- 
- a. PV Module
 - b. Solar panel
 - c. Photovoltaic array
 - d. Two of the choices above



photovoltaic (PV) array

- composed of PV modules connected in combination of series and parallel connections, which convert energy from the sun into electrical energy

PV module or solar panel

- assembly of photovoltaic cells mounted in a frame that uses sunlight as a source of energy to generate a DC electricity

Source: PNS/BAFS 324:2021



A solar power company wants to install solar panels on a building's rooftop. However, the site experiences cloudy weather frequently, causing sunlight to scatter.

What type of solar radiation will the panels primarily rely on?

- a. Scattered radiation
- b. Diffused radiation
- c. Direct solar radiation
- d. Reflected solar radiation



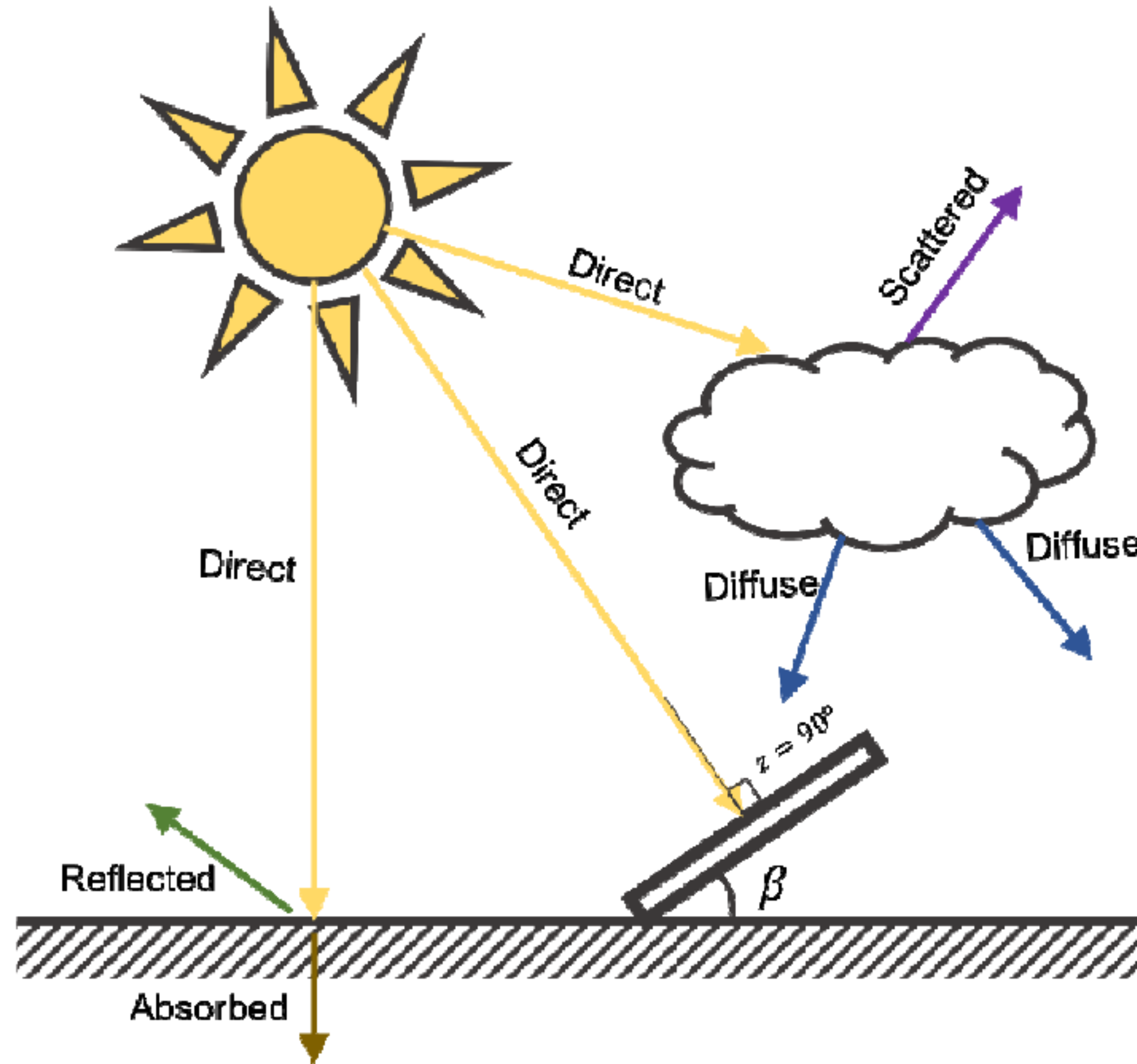
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- 
- 
- **Global Solar Radiation** – The total solar radiation received on a surface, including both direct and diffuse components.
 - **Direct Solar Radiation (Beam Radiation)** – The solar radiation that travels in a straight line from the Sun to the surface without scattering.
 - **Scattered Radiation** – A component of solar radiation that has been redirected in multiple directions by molecules, aerosols, or clouds in the atmosphere.

- 
- 
- **Diffused radiation** – This is the part of scattered sunlight that reaches the Earth's surface indirectly, without coming in a straight line from the Sun.
 - **Albedo Radiation** – Solar radiation that is reflected from the Earth's surface back into the atmosphere
 - **Reflected Solar Radiation** – Radiation that bounces off surfaces like the ground, buildings, or water, contributing to the total energy in the area.





It measures the total (or global) solar radiation.

- 
- a. Solar collector
 - b. Pyranometer
 - c. Pyrliometer
 - d. Spectroradiometer



It measures the total (or global) solar radiation.

- 
- a. Solar collector
 - b. Pyranometer**
 - c. Pyrheliometer
 - d. Spectroradiometer

- 
- 
1. **Pyranometer** – Measures global solar radiation (direct and diffuse).
 2. **Pyrheliometer** – Measures direct solar radiation (beam radiation).
 3. **Solar Power Meter** – Measures output of solar panels.
 4. **Sunshine Recorder** – Measures duration of sunshine.
 5. **Photovoltaic (PV) Cell Tester** – Tests performance of solar cells and modules.
 6. **Data Logger** – Records solar energy data over time.
 7. **Spectroradiometer** – Measures spectral distribution of solar radiation.
 8. **Radiometer** – Measures intensity of solar radiation.



Which battery is designed for slow charging and slow discharging?

- a. Maintenance-free battery
- b. Automotive battery
- c. Solar battery
- d. All of the above



Which battery is designed for slow charging and slow discharging?

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Kind of battery useful in solar-powered energy systems.

- a. Lead-acid battery
- b. Automotive battery
- c. Solar battery
- d. All of the above



Kind of battery useful in solar-powered energy systems.

- a. Lead-acid battery
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Material used in solar cells.

- a. Silicon
- b. Boron
- c. Equal silicon and boron
- d. Silicon with little boron



Material used in solar cells.

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- d. Silicon with little boron



In a sunny 12:00 noon in the Philippines, what is the likely voltage reading of solar pump without battery when load is in turned on?

- a. 20–22 V
- b. 16–19 V
- c. 14–15 V
- d. 11–13 V



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In a sunny 12:00 noon in the Philippines, what is the likely voltage reading of solar battery charger when no battery is being charged?

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- b. 16–19 V
- c. 14–15 V
- d. 11–13 V



In a sunny 12:00 noon in the Philippines, what is the likely voltage reading of solar battery charger when no battery is being charged?

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At 8:00 P.M. in the Philippines, what is the likely voltage reading of a solar home system when all loads are turned on?

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- b. 16–19 V
- c. 14–15 V
- d. 11–13 V



At 8:00 P.M. in the Philippines, what is the likely voltage reading of a solar home system when all loads are turned on?

- a. 20–22 V
- b. 16–19 V
- c. 14–15 V
- d. 11–13 V



If a coconut tree shades half of solar module, which condition is true?

- a. Zero voltage
- b. Low voltage
- c. Zero amperage
- d. Low amperage



If a coconut tree shades half of solar module, which condition is true?

- a. Zero voltage
- b. Low voltage
- c. Zero amperage
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A decorative background pattern consisting of a grid of hexagons, some of which are filled with a light gray color, creating a honeycomb-like effect.

What is the peak solar power generation per charging channel of a solar battery charging station equipped with 24 units of 150 watt-peak 24-volt solar module? The system is divided into 6 charging channels, each channel has 4 modules connected in parallel. Assume 70% overall efficiency.

- a. 600 W
- b. 540 W
- c. 420 W
- d. 300 W

What is the peak solar power generation per charging channel of a solar battery charging station equipped with 24 units of 150 watt-peak 24-volt solar module? The system is divided into 6 charging channels, each channel has 4 modules connected in parallel. Assume 70% overall efficiency.

Calculate the total power of the solar modules in one channel:

Total power per channel = # of modules x Power

Total power per channel = 4 modules x 150 W/module

Total power per channel = 600 W

Peak power generation per channel = 600 W x 0.70 = **420 W**

A decorative background pattern consisting of a grid of hexagons, some of which are filled with a light gray color, creating a honeycomb-like effect.

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- a. 600 W
- b. 540 W
- c. 420 W
- d. 300 W

A decorative background pattern of light gray hexagons is visible at the top of the slide, spanning across the width of the content area.

What is the system voltage of a solar pumping system equipped with 2 units of 75 watt-peak solar modules? Each solar module has 4.16 amperes maximum, open circuit voltage of 21 volts maximum and nominal voltage of 12 volts. Modules are connected in series.

- a. 6 volts
- b. 12 volts
- c. 24 volts
- d. 36 volts

What is the system voltage of a solar pumping system equipped with 2 units of 75 watt-peak solar modules? Each solar module has 4.16 amperes maximum, open circuit voltage of 21 volts maximum and nominal voltage of 12 volts. Modules are connected in series.

Voltage = (2 units)(12 nominal volts)

System voltage of a solar pumping system = **24 volts**

A decorative background element consisting of a light gray hexagonal grid pattern, partially visible at the top of the slide.

What is the system voltage of a solar pumping system equipped with 2 units of 75 watt-peak solar modules? Each solar module has 4.16 amperes maximum, open circuit voltage of 21 volts maximum and nominal voltage of 12 volts. Modules are connected in series.

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In an aerogenerator exceeding 100 kw capacity, which part is responsible in converting rotational movement to electricity?

- a. Rotor
 - b. Turbine
 - c. Alternator
 - d. Governor
- 



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- 



In multi-bladed windpumps, which part is responsible in converting rotational movement to reciprocating action?

- A. Rotor
- B. Connecting rod
- C. Reciprocating pump
- D. Crank arm



In multi-bladed windpumps, which part is responsible in converting rotational movement to reciprocating action?

- A. Rotor
- B. Connecting rod**
- C. Reciprocating pump
- D. Crank arm



This vertical axis windmill has a very low tip speed ratio and power coefficient of less than 0.2. The rotor is usually constructed using a cylindrical tank cut axially into two and installed vertically face to face at an offset position.

- 
- a. Savonius
 - b. Darrieus
 - c. Multibladed
 - d. Kaplan



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- 
- a. Savonius
 - b. Darrieus
 - c. Multibladed
 - d. Kaplan



If typical overall efficiency of savonius windmills is 7–10%, what is the typical overall efficiency of aerogenerators?

- A. 7–10%
- B. 15–17%
- C. 18–21%
- D. 90–95%



If typical overall efficiency of savonius windmills is 7–10%, what is the typical overall efficiency of aerogenerators?

- A. 7–10%
- B. 15–17%
- C. 18–21%
- D. 90–95%



What component of force is used in Darrieus windmills?

- a. Drag Force
 - b. Lift Force
 - c. Side Force
 - d. Centrifugal Force
- 



What component of force is used in Darrieus windmills?

- a. Drag Force
 - b. Lift Force**
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 - d. Centrifugal Force
- 



What component of force
is used in Savonius
windmills?

- a. Drag Force
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- 



What component of force
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- a. Drag Force
 - b. Lift Force
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- 



The force that acts
perpendicular to the
direction of motion.

- 
- a. Drag Force
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The force that acts
perpendicular to the
direction of motion.

- 
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 - b. Lift Force**
 - c. Side Force
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What component of force
is used in multi-bladed
windmills?

- a. Lift force
 - b. Drag force
 - c. Side force
 - d. Drop force
- 



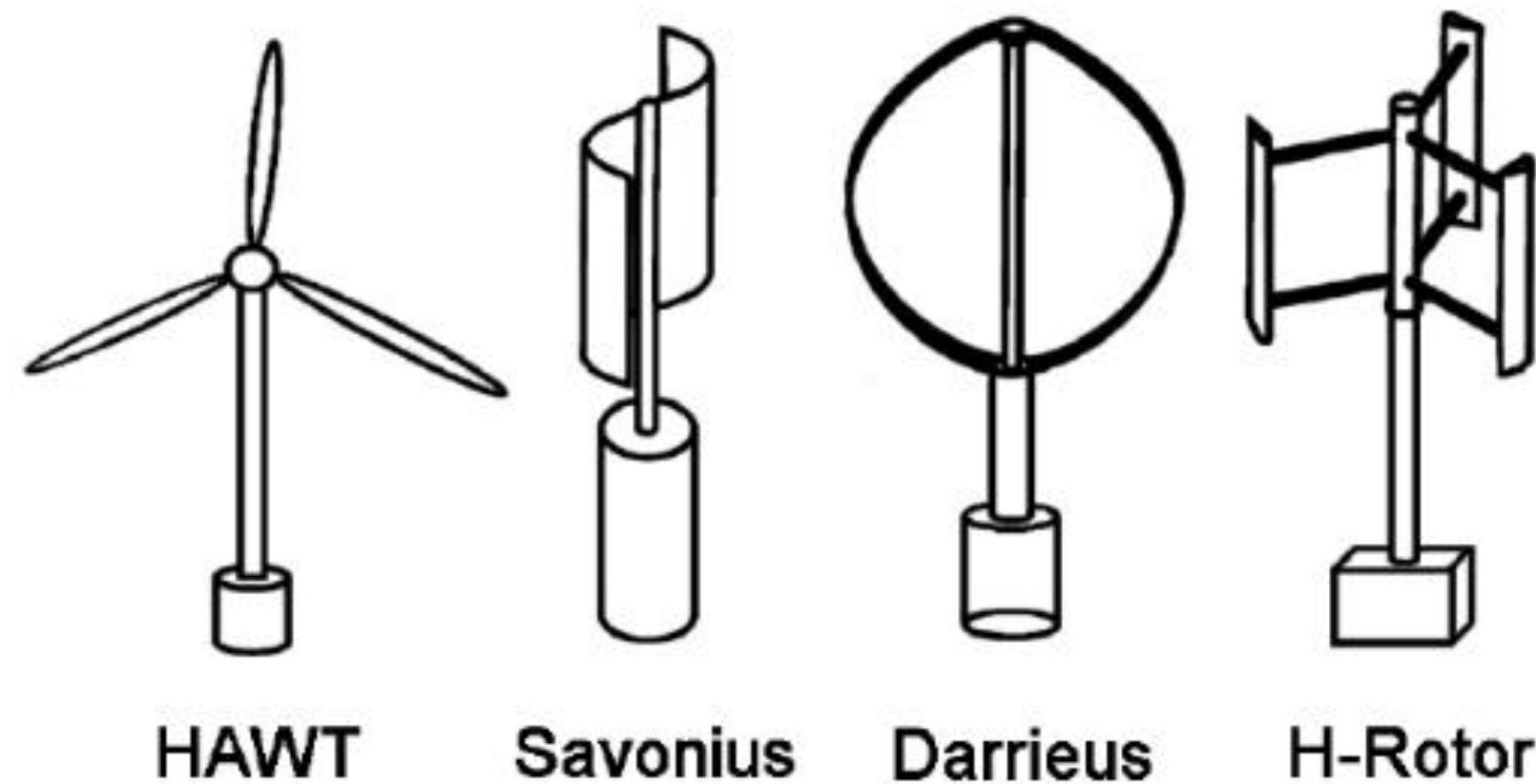
What component of force
is used in multi-bladed
windmills?

- a. Lift force
- b. Drag force
- c. Side force
- d. Drop force



Lift Force – The force that acts perpendicular to the direction of motion, causing an object to rise.

Drag Force – The force that acts parallel and opposite to the direction of motion, resisting movement.



1. **Savonius** – Drag-based VAWT, good for turbulent wind.
2. **H-Rotor** – VAWT with an "H" shape, operates in various wind directions.
3. **HAWT** – Horizontal rotor shaft, highly efficient, common in wind farms.
4. **Darrieus** – Lift-based VAWT, efficient at higher wind speeds.



The type of pump is appropriate for multi-bladed windmills.

- a. Hydraulic
- b. Pneumatic
- c. Reciprocating
- d. Diaphragm



The type of pump is appropriate for multi-bladed windmills.

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What wind energy resource is located about 10–16 km from Earth surface with velocity over 150 kph?

- A. Terrestrial wind
- B. Extra-terrestrial wind
- C. Jet stream
- D. High wind



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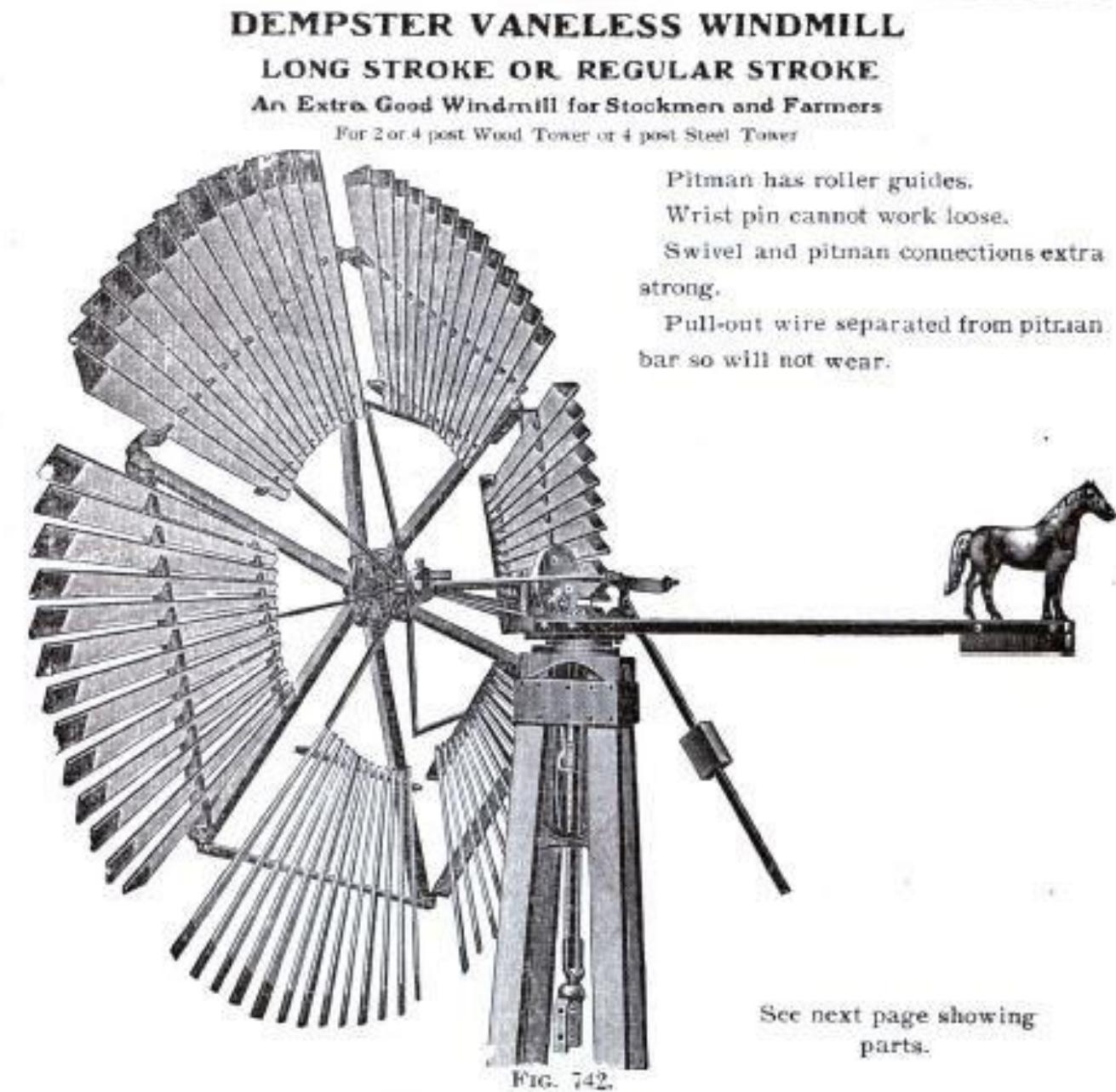


What tail type of multi-bladed windpump is more effective for windy areas?

- a. Hinged side vane
- b. Ecliptic
- c. Manual folding
- d. No tail

What tail type of multi-bladed windpump is more effective for windy areas?

- a. Hinged side vane
- b. Ecliptic
- c. Manual folding
- d. No tail





In windmill design, what should be the average rotor blade twist angle in degrees to have a tip speed ratio of 1:1?

- a. 30
- b. 45
- c. 60
- d. 75



In windmill design, what should be the average rotor blade twist angle in degrees to have a tip speed ratio of 1:1?

- a. 30
- b. 45**
- c. 60
- d. 75

With a wind velocity of 8 m/s, how much power can a 3 m diameter wind turbine theoretically generate?

- a. 2126 W b. 2162 W c. 2621 W d. 2261 W

How much power is available if it will be used for maximum mechanical energy?

- a. 1000.6 W b. 1266.4 W c. 1334.5 W d. 1522.8 W



Theoretical Power = $\frac{1}{2}\rho AV^3$ where $\rho=1.25 \text{ kg/m}^3$

Actual power:

$P \text{ (watts)} = 0.245AV^3$ (For mechanical energy conversion)

$P \text{ (watts)} = 0.10AV^3$ (For electrical conversion)

Available Power:

$P = \text{Theoretical } P \times \text{Power Coefficient}$

Power Coeff:

0.59 (For mechanical energy conversion)

0.20 (For electrical conversion)



With a wind velocity of 8 m/s, how much power can a 3 m diameter wind turbine theoretically generate? How much power is available if it will be used for maximum mechanical energy?

$$\text{Theoretical Power} = \frac{1}{2} \rho A V^3$$

$$\text{Theoretical Power} = \frac{1}{2} (1.25 \text{ kg/m}^3) \left(\frac{\pi (3 \text{ m})^2}{4} \right) (8 \text{ m/s})^3$$

$$\text{Theoretical Power} = 2261 \text{ Watts}$$

How much power is available if it will be used for maximum mechanical energy?

Max Available Power = Theoretical P x Power Coeff

Max Available Power = 2261 W x 0.59

Max Available Power = 1334.5 W



With a wind velocity of 8 m/s, how much power can a 3 m diameter wind turbine theoretically generate?

- a. 2126 W b. 2162 W c. 2621 W **d. 2261 W**

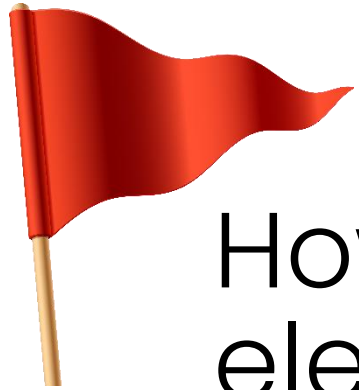
How much power is available if it will be used for maximum mechanical energy?

- a. 1000.6 W b. 1266.4 W **c. 1334.5 W** d. 1522.8 W



With a wind velocity of 10 m/s, how much power can a π m diameter wind turbine theoretically generate?

- a. 4300 W b. 4565 W c. 4845 W d. 4997 W



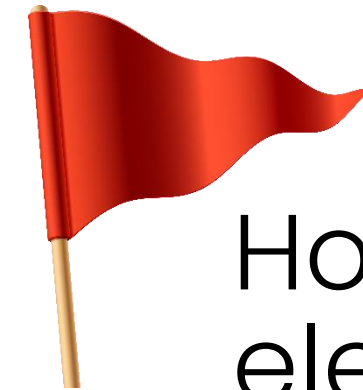
How much power is available if it will be used for maximum electrical energy?

- a. 689.95 W b. 896.59 W c. 968.95 W d. 986.59 W



With a wind velocity of 10 m/s, how much power can a π m diameter wind turbine theoretically generate?

- a. 4300 W b. 4565 W c. 4845 W d. 4997 W



How much power is available if it will be used for maximum electrical energy?

- a. 689.95 W b. 896.59 W c. 968.95 W d. 986.59 W

With a wind velocity of 10 m/s, how much power can a π m diameter wind turbine theoretically generate?

$$\text{Theoretical Power} = \frac{1}{2} \rho A V^3$$

$$\text{Theoretical Power} = \frac{1}{2} (1.25 \text{ kg/m}^3) \left(\frac{\pi (2\pi \text{ m})^2}{4} \right) (10 \text{ m/s})^3$$

$$\text{Theoretical Power} = 4844.73 \text{ Watts}$$

How much power is available if it will be used for maximum mechanical energy?

Max Available Power = Theoretical P x Power Coeff

Max Available Power = 4844.73 x 0.20

Max Available Power = 968.95 W



A multi-sail type windmill for pumping water has a diameter of 6 ft and the wind velocity is 10 mph. Estimate the actual power of this windmill.

- a. 0.06
 - b. 0.77
 - c. 5.45
 - d. 2.42
- 



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A multi-sail type windmill for pumping water has a diameter of 6 ft and the wind velocity is 10 mph. Estimate the actual power of this windmill.

$$D = 6 \text{ ft} \times \frac{1 \text{ m}}{3.28 \text{ ft}} = 1.83 \text{ m}$$

$$V = 10 \frac{\text{miles}}{\text{hour}} \times \frac{1.61 \text{ km}}{1 \text{ mile}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{3600 \text{ seconds}} = 4.47 \text{ m/s}$$

$$P = 0.245 \left(\frac{\pi (1.83 \text{ m})^2}{4} \right) (4.47 \text{ m/s})^3$$

$$P = 57.479 \text{ W or } 0.057 \text{ kW}$$



Given the rotational speed of turbine is 150 rpm, calculate the tip speed ratio for a 3m diameter wind turbine and with a wind velocity of 8 m/s.

- a. 1.43
 - b. 2.01
 - c. 2.95
 - d. 3.15
- 


$$\text{TSR} = \frac{\text{Blade tip speed}}{\text{Wind speed}} = \frac{\pi D N}{V}$$

Given the rotational speed of turbine is 150 rpm, calculate the tip speed ratio for a 3m diameter wind turbine and with a wind velocity of 8 m/s.

$$\text{TSR} = \frac{\text{Blade tip speed}}{\text{Wind speed}} = \frac{\pi D N}{V}$$

$$\text{TSR} = \frac{\pi(3 \text{ m})\left(150 \frac{\text{rev}}{\text{min}}\right)\left(\frac{1 \text{ min}}{60 \text{ sec}}\right)}{\left(8 \frac{\text{m}}{\text{s}}\right)} = 2.95$$



Given the rotational speed of turbine is 150 rpm, calculate the tip speed ratio for a 3m diameter wind turbine and with a wind velocity of 8 m/s.

- a. 1.43
 - b. 2.01
 - c. 2.95**
 - d. 3.15
- 



Given the rotational speed of turbine is 320 rpm, calculate the tip speed ratio for a wind turbine that theoretically generates a power of 4,177.3 W and with a wind velocity of 6 m/s.

- a. 14.5
- b. 15.98
- c. 16.09
- d. 17.53



Given the rotational speed of turbine is 320 rpm, calculate the tip speed ratio for a wind turbine that theoretically generates a power of 4,177.3 W and with a wind velocity of 6 m/s.

$$\text{Theoretical Power} = \frac{1}{2} \rho A V^3$$

$$4,177.3 \text{ W} = \frac{1}{2} (1.25 \text{ kg/m}^3) \left(\frac{\pi (D \text{ m})^2}{4} \right) (6 \text{ m/s})^3$$

$$D = 6.2768 \text{ m}$$

$$\text{TSR} = \frac{\pi D N}{V} = \frac{\pi (6.2768 \text{ m}) \left(320 \frac{\text{rev}}{\text{min}} \right) \left(\frac{1 \text{ min}}{60 \text{ sec}} \right)}{6 \text{ m/s}}$$

$$\text{TSR} = 17.53$$



Given the rotational speed of turbine is 320 rpm, calculate the tip speed ratio for a wind turbine that theoretically generates a power of 4,177.3 kW and with a wind velocity of 6 m/s.

- a. 14.5
- b. 15.98
- c. 16.09
- d. 17.53**



While the windmill is in operation, what efficiency can exceed 100%?

- a. Volumetric
- b. Transmission
- c. Rotor
- d. All of the above



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In wind farm development, which should be done first?

- a. Rotor
- b. Energy demand survey
- c. Transmission
- d. Wind velocity measurement



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To have bigger water displacement volume in windpumps, which should be increased?

- a. Connecting rod length
- b. Rotor diameter
- c. Crank arm
- d. Well diameter



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Under Philippine classification,
what is the maximum power
rating for microhydro?

- a. 25 kw
 - b. 50 kw
 - c. 75 kw
 - d. 100 kw
- 



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- a. 25 kw
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A 260 kw hydro power plant is classified as

- a. Microhydro
- b. Minihydro
- c. Small hydro
- d. Big hydro



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HYDRO POWER PLANT CLASSIFICATION BASED ON GENERATED POWER

1. Pico Hydro ($P < 5 \text{ kW}$)
2. Micro Hydro ($5 \text{ kW} \leq P \leq 100 \text{ kW}$)
3. Mini Hydro ($100 \text{ kW} \leq P \leq 10 \text{ MW}$)
4. Small Hydro ($10 \text{ MW} \leq P \leq 25 \text{ MW}$)
5. Medium Hydro ($25 \text{ MW} \leq P \leq 100 \text{ MW}$)
6. Large Hydro ($P > 100 \text{ MW}$)



Which type of turbine operates based on converting the entire pressure energy of the fluid into kinetic energy before striking the blades?

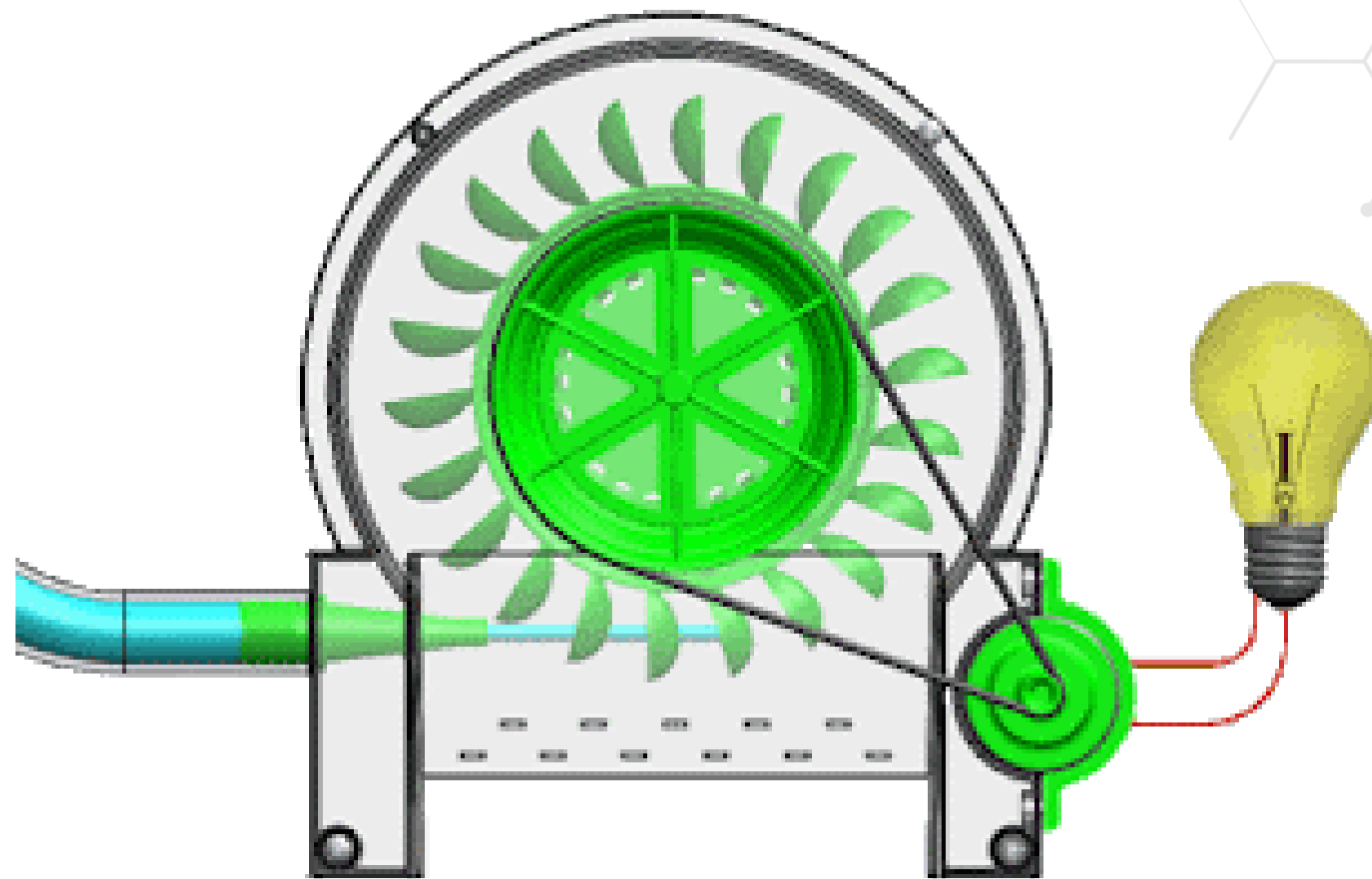
- 
- a. Impulse turbine
 - b. Reaction turbine
 - c. Cross-flow turbine
 - d. Mixed-flow turbine



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An example of an impulse turbine is the **Pelton Wheel Turbine**. This turbine is commonly used in hydroelectric plants where high-pressure water jets hit the turbine buckets to generate power. The water strikes the blades at high velocity, and there is no change in pressure as it moves through the turbine.





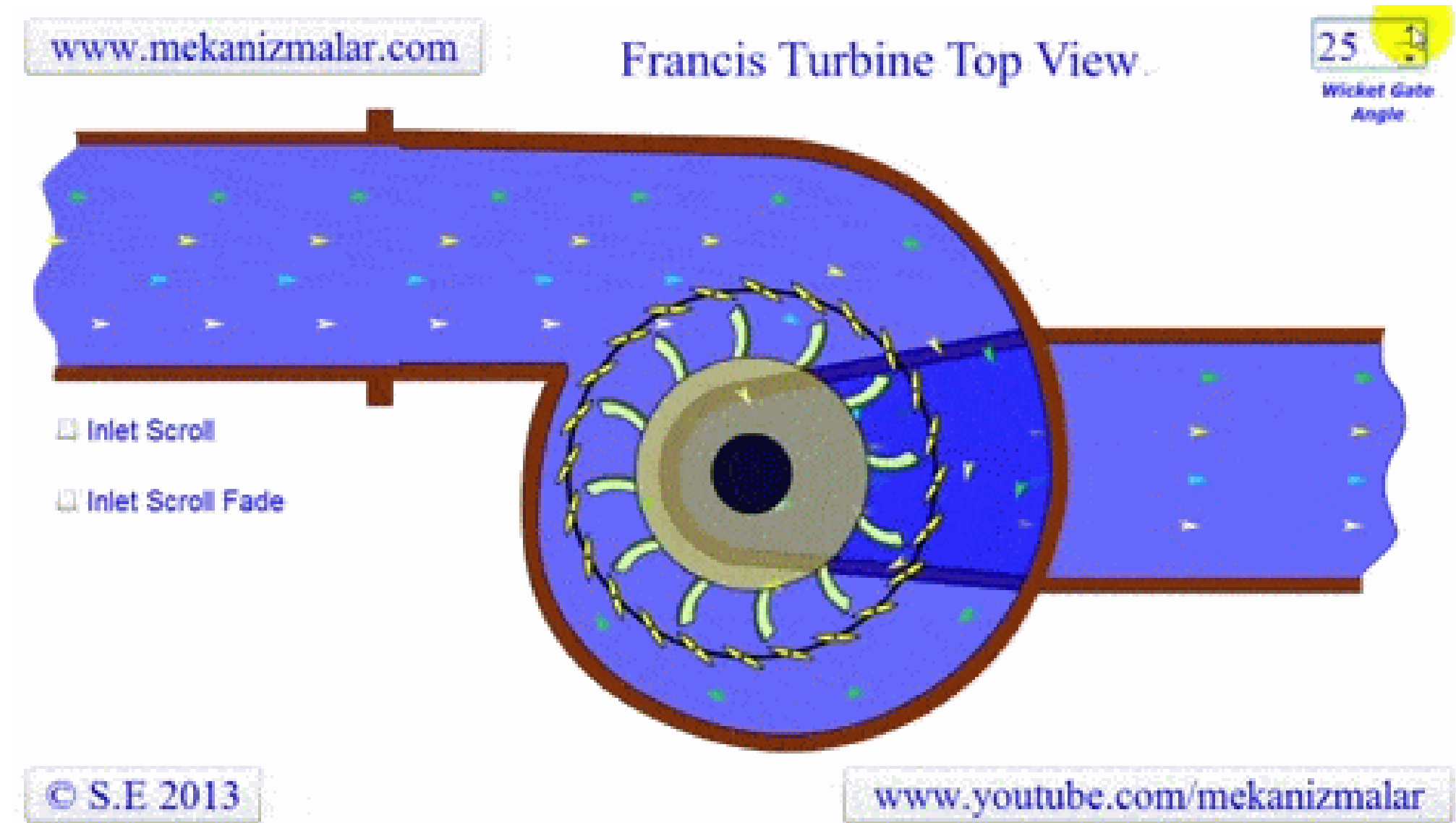
Which type of turbine works on the principle of a gradual pressure drop as fluid flows through the blades, with both pressure and kinetic energy contributing to the output?

- 
- a. Impulse turbine
 - b. Reaction turbine
 - c. Cross-flow turbine
 - d. Francis turbine



Which type of turbine works on the principle of a gradual pressure drop as fluid flows through the blades, with both pressure and kinetic energy contributing to the output?

- 
- a. Impulse turbine
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 - d. Francis turbine



How Francis Turbine (Reaction Type) Works?

Pressure + Kinetic Energy → Spins the Runner → Mechanical Energy → Generator → Electricity



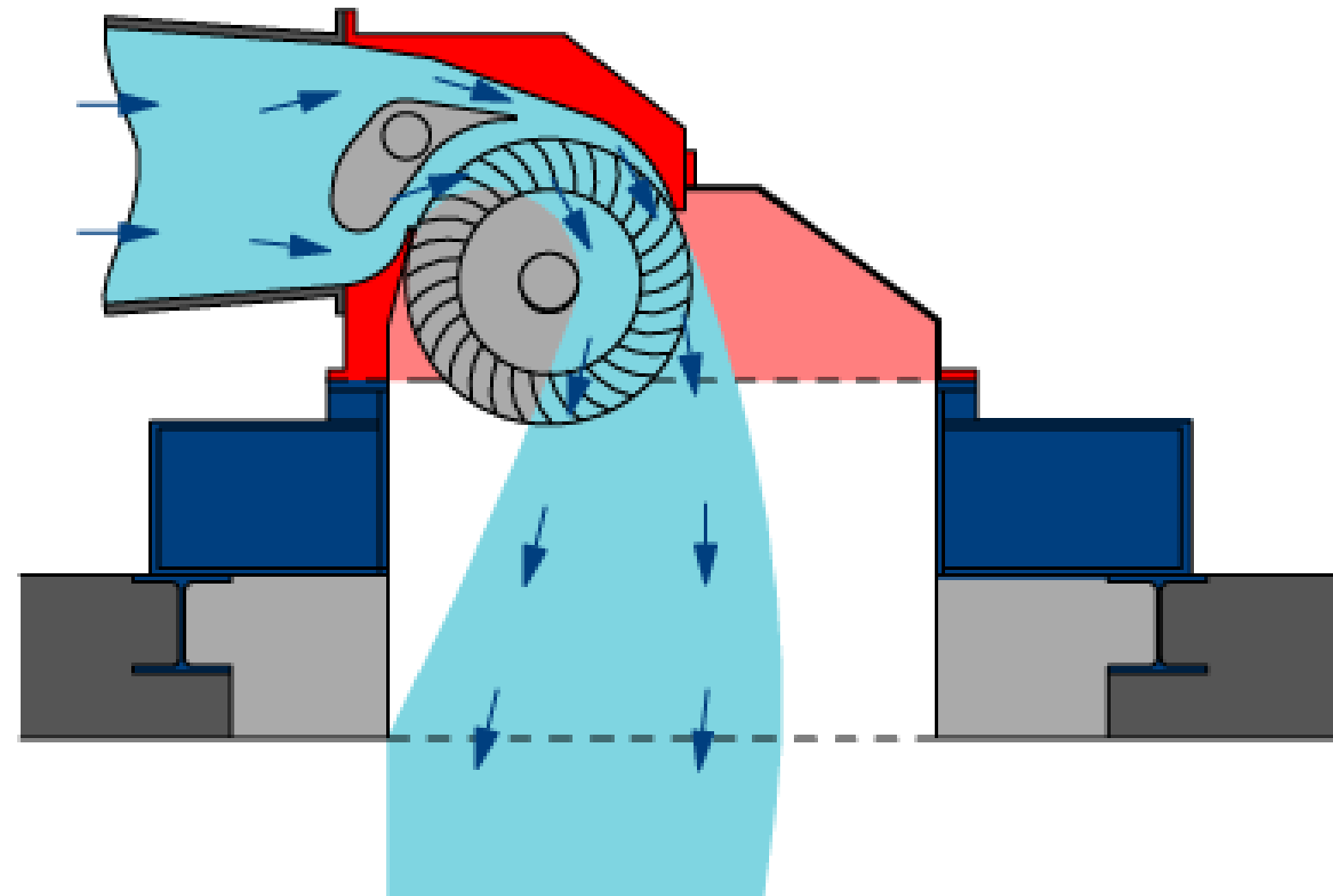
Banki-Michel Turbine is an example of what type of turbine?

- 
- a. Impulse turbine
 - b. Reaction turbine
 - c. Cross-flow turbine
 - d. Francis turbine



Banki-Michel Turbine is an example of what type of turbine?

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A cross-flow turbine is a type of hydroelectric turbine used primarily for small-scale power plants with low to medium heads (waterfall heights). It is unique because the water flows perpendicularly (across) the turbine blades in two stages—entering on one side and exiting on the opposite side, passing through the runner blades twice.

Determine the minimum head requirements for a water turbine to generate 8 hp if it has a conversion efficiency of 50% and the discharge rate from stream flow through pipe conduit is 2000 L/min to run the turbine.

- a. 73.08 m
- b. 36.53 m
- c. 73.08 ft
- d. 36.53 ft

Determine the minimum head requirements for a water turbine to generate 8 hp if it has a conversion efficiency of 50% and the discharge rate from stream flow through pipe conduit is 2000 L/min to run the turbine.

$$Q = \frac{2000\text{L}}{\text{min}} \times \frac{1\text{ min}}{60\text{ sec}} \times \frac{1\text{ m}^3}{1000\text{L}} = 0.0333\text{m}^3/\text{sec}$$

$$P = 8\text{ hp} \times \frac{746\text{ W}}{1\text{ hp}} = 5968\text{ W}$$

$$\text{Hydro Power Required} = 5968\text{ W} / 0.50 = 11,936\text{ W}$$

$$P = \rho g Q H$$

$$11,936\text{ W} = (1000\text{ kg/m}^3) \left(9.81 \frac{\text{m}}{\text{s}^2} \right) (0.0333\text{m}^3/\text{sec}) H$$

$$H = 36.53\text{ m}$$

Determine the minimum head requirements for a water turbine to generate 8 hp if it has a conversion efficiency of 50% and the discharge rate from stream flow through pipe conduit is 2000 L/min to run the turbine.

- a. 73.08 m
- b. 36.53 m**
- c. 73.08 ft
- d. 36.53 ft



Water falls at a rate of 100 L/sec at a height of 5 meters.
What is the theoretical power available?

- a. 2943 W
- b. 3.2 kW
- c. 4905 W
- d. 5.17 kW

Water falls at a rate of 100 L/sec at a height of 5 meters.
What is the theoretical power available?

$$Q = \frac{100\text{L}}{\text{sec}} \times \frac{1\text{ m}^3}{1000\text{L}} = 0.1\text{ m}^3/\text{sec}$$

$$P = \rho g Q H$$

$$P = (1000\text{ kg/ m}^3) \left(9.81 \frac{\text{m}}{\text{s}^2} \right) (0.1\text{ m}^3/\text{sec})(5\text{ m})$$

$$P = 4905\text{ W or } 4.9\text{ kW}$$



Water falls at a rate of 100 L/sec at a height of 5 meters.
What is the theoretical power available?

- a. 2943 W
- b. 3.2 kW
- c. 4905 W
- d. 5.17 kW



What is the power generated by a micro hydro power plant having a gross head of 60 m and reliable discharge of 80 lps. Assume typical efficiency.

- a. 23.52 kW
- b. 28.25 kW
- c. 47.09 kW
- d. Not enough information



What is the power generated by a micro hydro power plant having a gross head of 60 m and reliable discharge of 80 lps. Assume typical efficiency.

- a. 23.52 kW
- b. 28.25 kW
- c. 47.09 kW
- d. Not enough information

What is the power generated by a micro hydro power plant having a gross head of 60 m and reliable discharge of 80 lps. Assume typical efficiency.

$$Q = \frac{80\text{L}}{\text{sec}} \times \frac{1\text{ m}^3}{1000\text{L}} = 0.08\text{ m}^3/\text{sec}$$

Note Typical efficiency for hydro power plant = 60%

$$P = \rho g Q H \times \text{Eff}$$

$$P = (1000\text{ kg/ m}^3) \left(9.81 \frac{\text{m}}{\text{s}^2} \right) (0.08\text{ m}^3/\text{sec}) (60\text{ m}) (0.6)$$

$$P = 28,252.8\text{ W or } 28.25\text{ kW}$$



There are 1000 heads of chicken in a poultry farm. Estimate the kilos of that dung that can be produced by the chicken in one month.

- a. 750 kg
- b. 1000 kg
- c. 2000 kg
- d. 2250 kg

Estimated Daily Quantity of Available Manure

Animal	Manure Available (kg/day/animal)
Pigs	2.20 – 8.00
Cow	7.50 – 14.00
Buffalo	8.00 – 14.00
Horses	7.75 – 13.50
chicken	0.025 – 0.075

There are 1000 heads of chicken in a poultry farm. Estimate the kilos of that dung that can be produced by the chicken in one month.

Weight of dung = $0.075 \text{ kg/day-animal} \times 1000 \text{ animals} \times 30 \text{ days}$
Weight of dung = 2,250 kg



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- a. 750 kg
- b. 1000 kg
- c. 2000 kg
- d. 2250 kg



There are 1200 heads of buffalo in a farm. Estimate the kilos of that dung that can be produced by the buffalos in one month.

- a. 420, 000 kg
- b. 480, 000 kg
- c. 502, 000 kg
- d. 504, 000 kg

Gas Production Potential of Various Types of Manure in cu. m/kg

Manure	Retention Period			
	25 days	30 days	35 days	50 days
Pig	0.058	0.063	0.064	0.077
Cow	0.030	0.034	0.037	0.043
Buffalo	0.030	0.034	0.037	0.043
Horse	0.045	0.051	0.56	0.065
Chicken	0.060	0.065	0.069	0.078



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There are 1200 heads of buffalo in a farm. Estimate the kilos of that dung that can be produced by the buffalos in one month.

Weight of dung = $14 \text{ kg/day-animal} \times 1200 \text{ animals} \times 30 \text{ days}$

Weight of dung = 504,000 kg

A decorative background element consisting of a light gray hexagonal grid pattern, partially visible at the top of the slide.

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Determine the size of biogas digester required to produce biogas for livestock farm having 500 heads of cattle. The projected retention time is 25 days with feed material to water ratio of 1:1.5. (Cattle manure production = 14 kg/day-head; gas production = 0.03 m³/kg)

- a. 234.98 m³
- b. 437.5 m³
- c. 554.01 m³
- d. 665.23 m³

Determine the size of biogas digester required to produce biogas for livestock farm having 500 heads of cattle. The projected retention time is 25 days with feed material to water ratio of 1:1.5.

$$\text{Mass of manure} = 500 \text{ heads} \times \frac{14 \text{ kg}}{\text{day-head}} = 7000 \frac{\text{kg}}{\text{day}}$$

$$\text{Volume of manure} = \frac{\text{Mass}}{\text{Density of water}} = \frac{7000 \frac{\text{kg}}{\text{day}}}{1000 \text{ kg/m}^3} = 7 \text{ m}^3/\text{day}$$

Using 1:1.5 ratio,

Volume of slurry = Vol. of manure + Vol. of water

$$\text{Volume of slurry} = 7 \text{ m}^3/\text{day} + 1.5(7 \text{ m}^3/\text{day}) = 17.5 \text{ m}^3/\text{day}$$

$$\text{Volume of Digester} = 17.5 \text{ m}^3/\text{day} \times 25 \text{ days} = 437.5 \text{ m}^3$$

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What type of lubrication system uses a dipper attached to the connecting rod to distribute oil to engine components through mechanical turbulence?

- a. Pressure
 - b. Mist
 - c. Splash
 - d. Gravity
- 

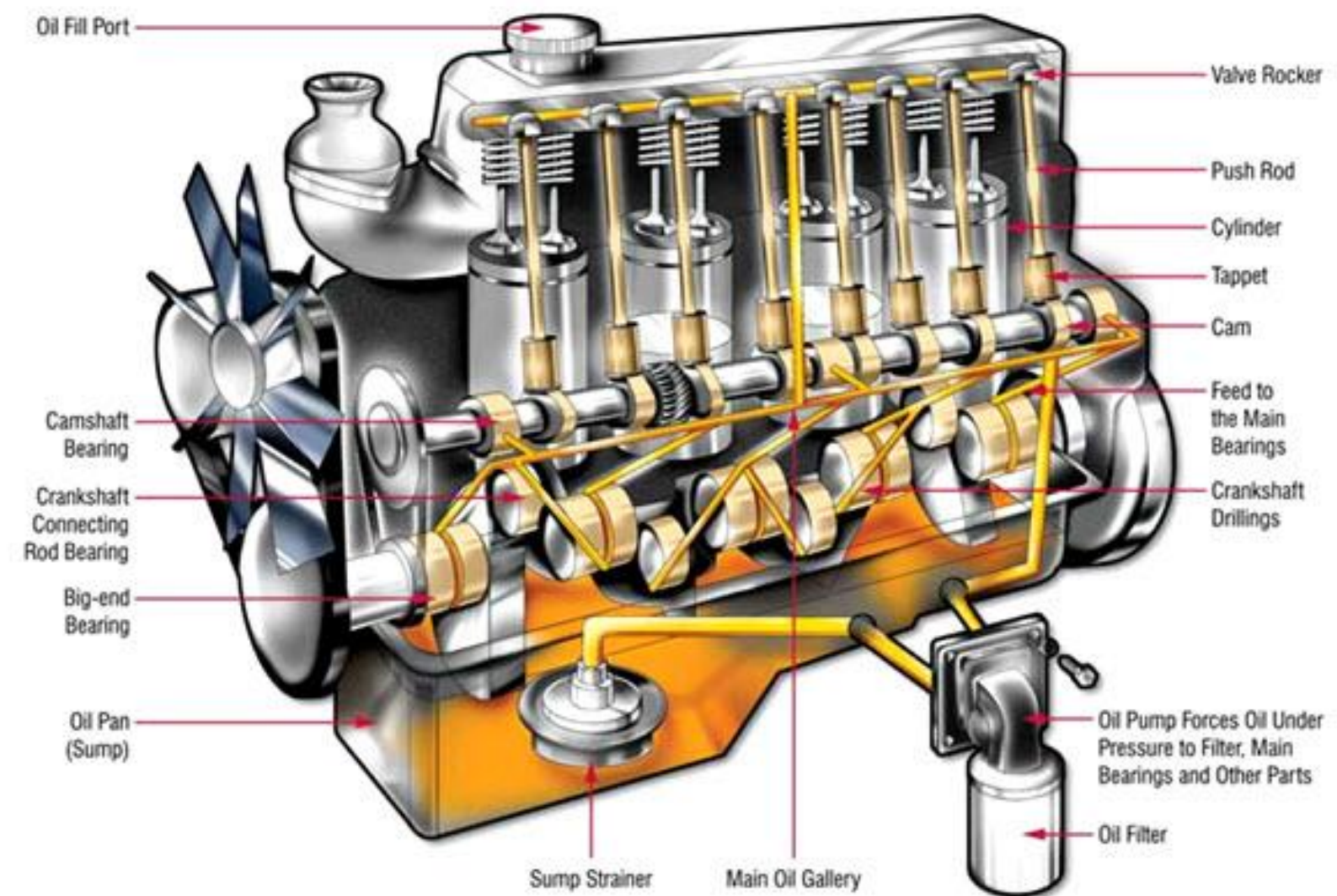


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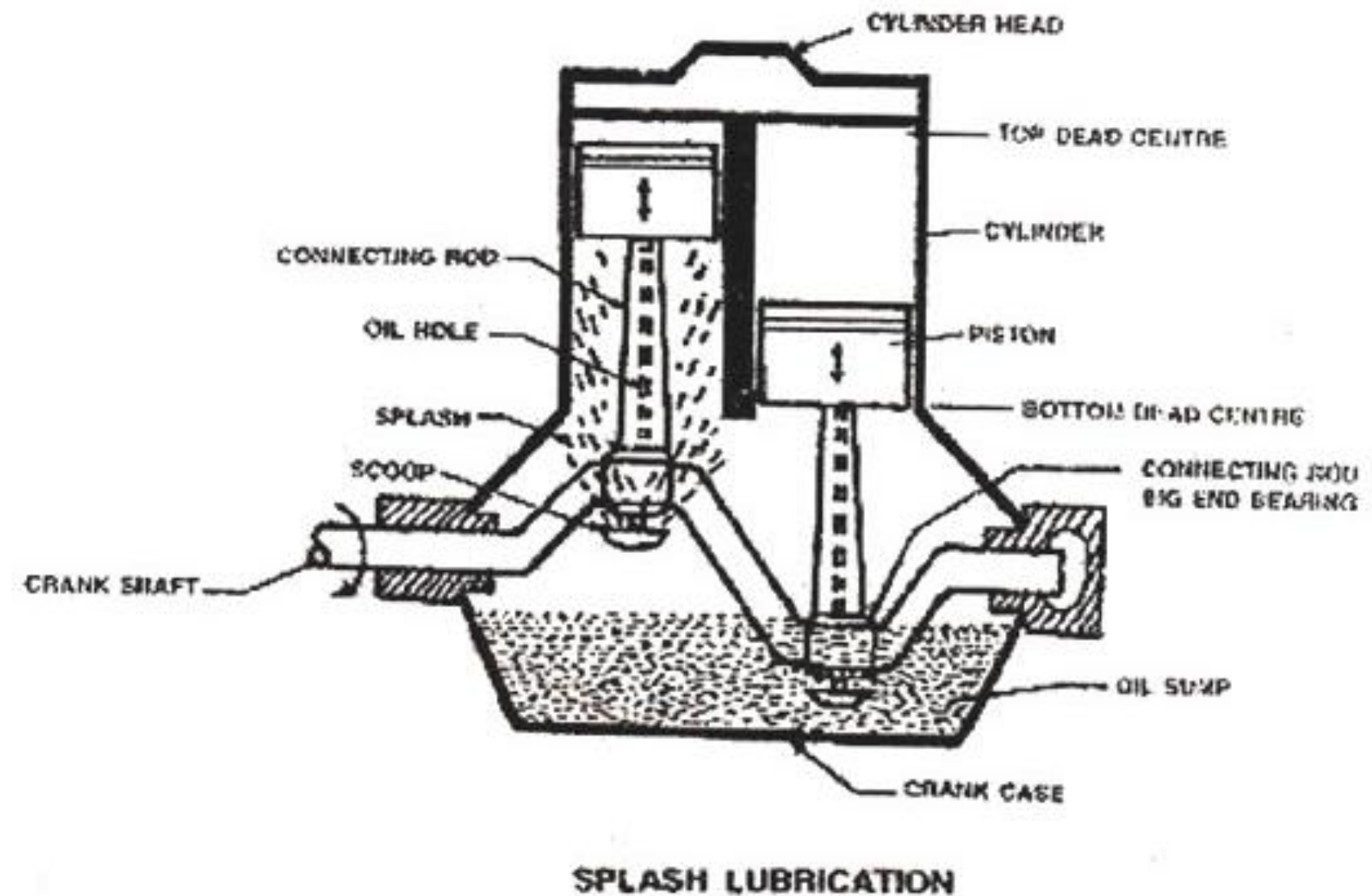
Force-Fed Lubrication

- Oil is pumped under pressure to critical engine parts through a network of oil passages.
- Delivers oil directly to components via oil pumps, ensuring consistent pressure and flow.



Splash Lubrication

- Oil is picked up by a dipper or similar device and splashed onto components by moving parts.
- Relies on mechanical movement and turbulence to distribute oil.





What property of oil refers to its resistance to flow and the separation of two surfaces?

- 
- a. Density
 - b. Viscosity
 - c. Flash Point
 - d. Pour Point



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What happens to the viscosity of lubricating oil at higher temperatures?

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- a. It increases
 - b. It remains constant
 - c. It decreases
 - d. It solidifies



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What does a high Viscosity Index (VI) indicate about an oil's viscosity with temperature changes?

- 
- a. Greater change in viscosity with temperature
 - b. Less change in viscosity with temperature
 - c. No effect of temperature on viscosity
 - d. Higher viscosity at all temperatures



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What is the significance of the pour point of lubricating oil?

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- a. It indicates the oil's viscosity at high temperatures.
 - b. It is the temperature below which the oil cannot flow.
 - c. It measures the oil's ability to resist oxidation.
 - d. It refers to the temperature at which oil ignites.



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Why is corrosion resistance important for lubricating oils in engines?

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- a. To prevent oil from vaporizing
 - b. To ensure oil remains clean
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 - d. To maintain oil viscosity

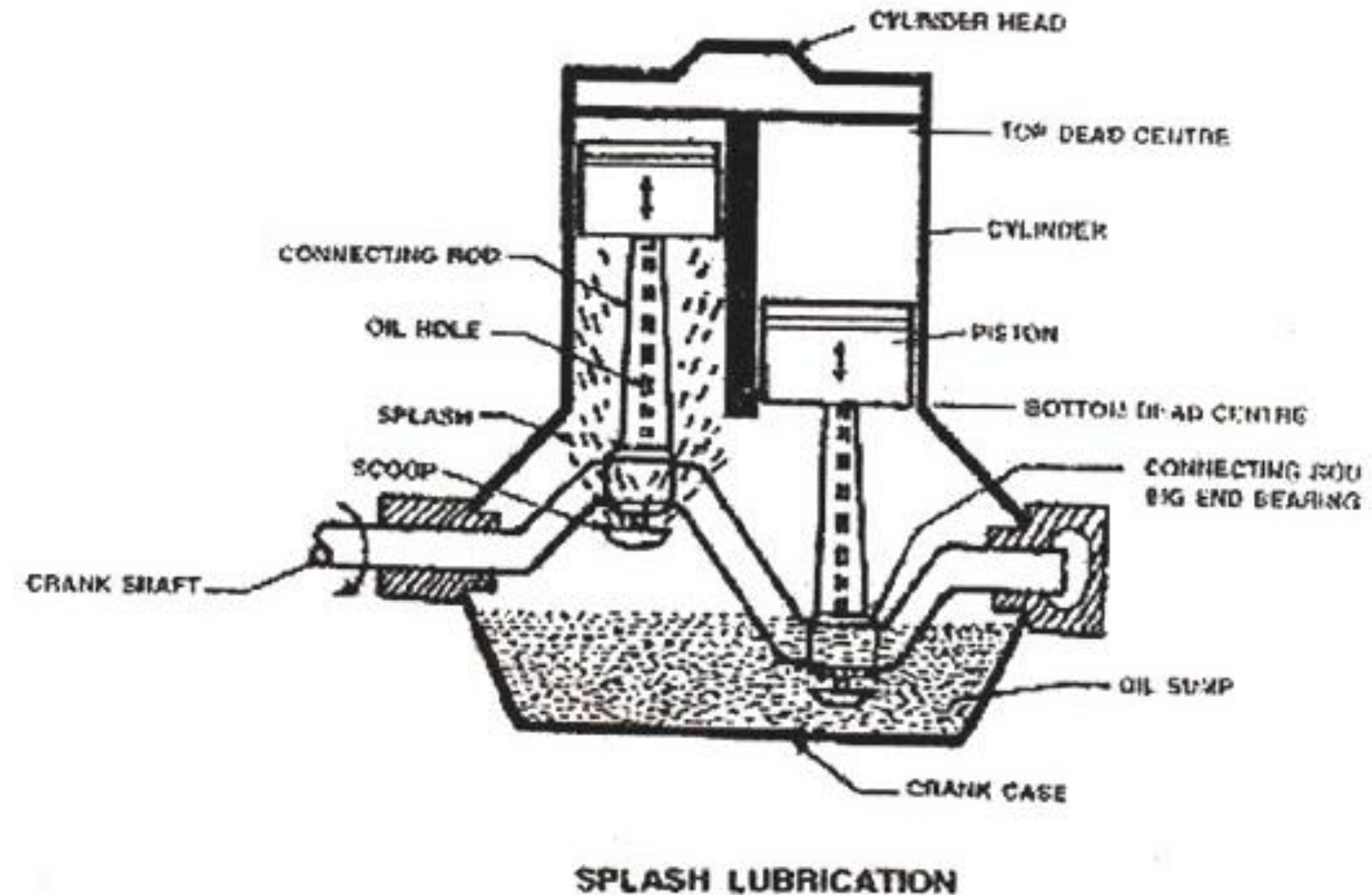


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PROPERTIES OF LUBRICANTS

1. **Viscosity** – The measure of a fluid's resistance to flow, indicating its ability to separate moving surfaces in an engine.
2. **Clean and Stable** – The quality of lubricating oils to remain free of contaminants and maintain their chemical properties under varying temperature conditions.
3. **Pour Point** – The lowest temperature at which oil can flow; below this temperature, the oil cannot perform its lubricating function.
4. **Flash Point** – The temperature at which oil vapors can ignite, indicating the oil's flammability and safety during operation.
5. **Corrosion Resistance** – The ability of lubricating oils to prevent damage and wear to engine components caused by chemical reactions and moisture.